Forestry Impacts on BC Pacific Salmon

Musgamagw Dzawada'enuxw territory watersheds

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Contents

Introduction

Logging has impacted many rivers in BC that support salmon. Using historical data (1950-2012) of pink salmon abundance in 417 rivers, chum salmon abundance in 374 rivers in BC, and of logging disturbance in these rivers, we estimated the impact that logging has on salmon recruitment (number of salmon produced in the next generation) using a statistical model. The two metrics of logging disturbance that we used are:

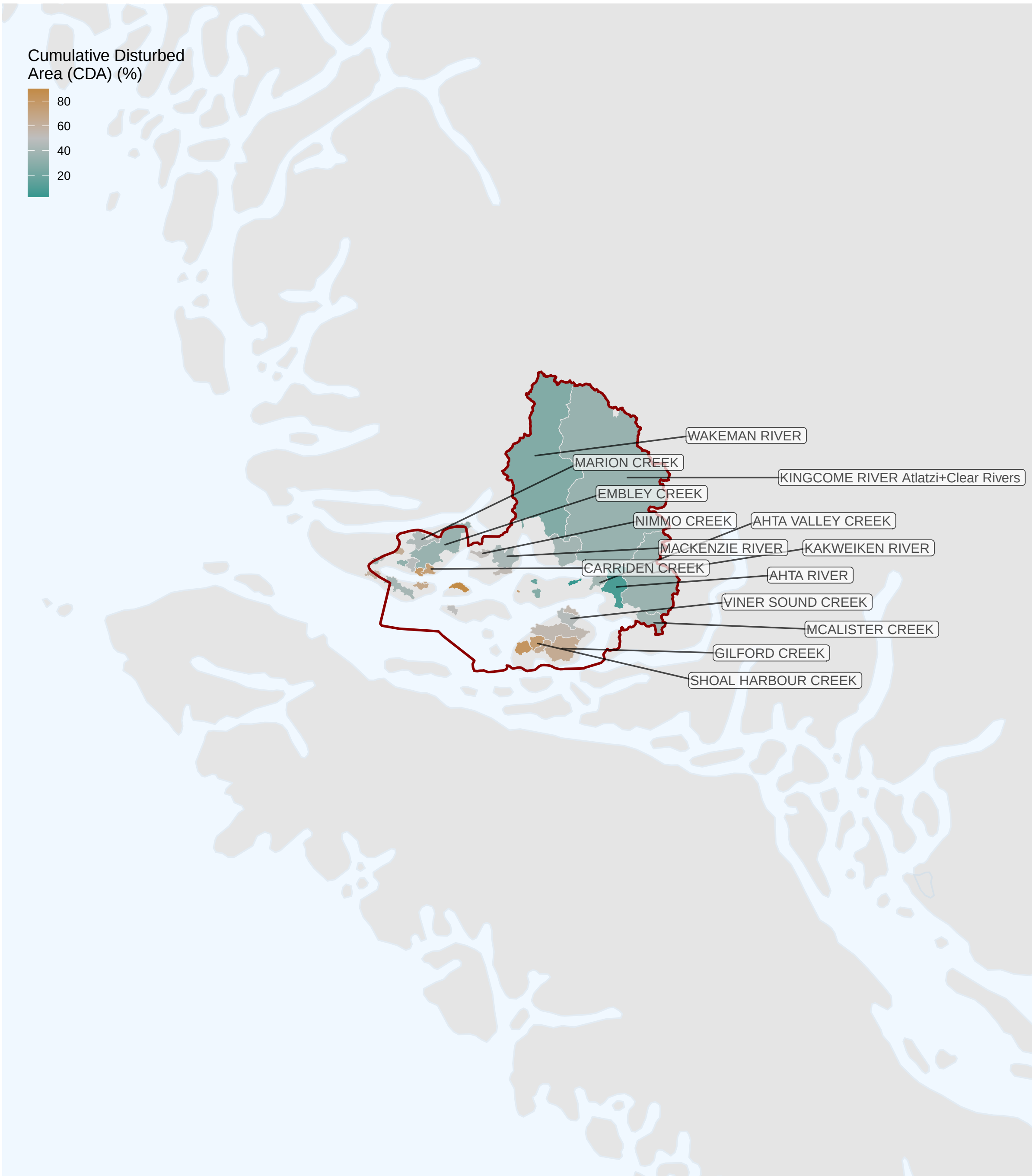
1) Cumulative Disturbed Area (CDA) - the percent of area around a river that has been disturbed by human activities or natural disturbances without assuming any recovery within the time period of our dataset (62 years).
2) Equivalent Clearcut Area (ECA) - proportion of area that has been clearcut around a river after accounting for an expected hydrologic recovery. This metric assumes that the forest is close to 100% recovered roughly 50 years post-logging.

In addition to logging disturbance, we also used two variables to account for ocean conditions:

3) Sea-Surface Temperature (SST) - the average sea-surface temperature corresponding to the time when the salmon first enters the ocean, which is known to have variable effects on salmon recruitment.
4) North Pacific Gyre Oscillation (NPGO) - index of primary productivity in the ocean, which is known to have a positive effect on salmon recruitment.

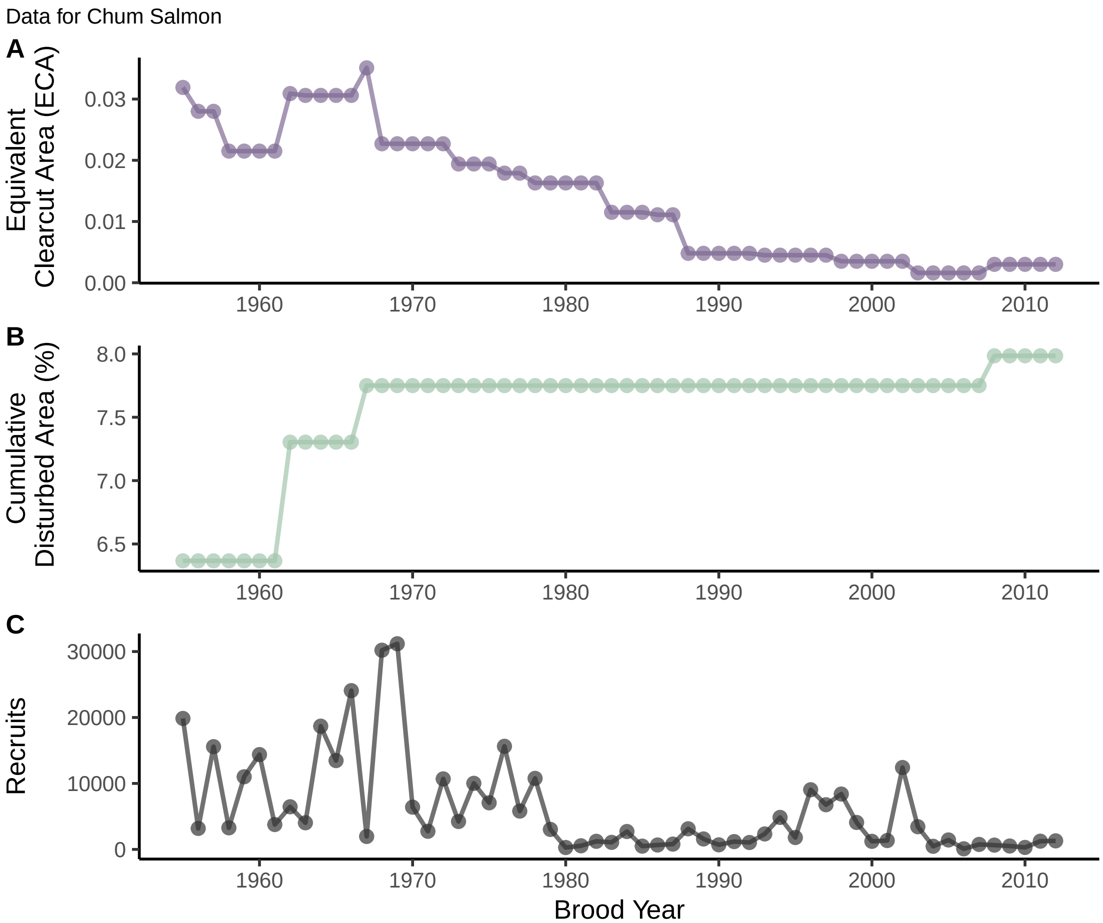
In this document, we have used the 4 coastwide models (with CDA for chum, with ECA for chum, with CDA for pink, and with ECA for pink) to estimate the impact of logging on some of the rivers within MD territory for which we had data. For each of these rivers, we have plotted the data of logging disturbance as given by the two metrics, ECA and CDA, and the number of salmon recruits with time. We then plotted the results of the model for that river A) the probability distributions of the effect of CDA, NPGO, and SST on salmon recruitment for that river (probability distributions with the peak to the left of 0 indicate a negative effect on salmon recruitment), B) the predicted spawner-recruit relationship for two different levels of logging disturbance experienced by the river, C) the predicted change in recruitment for different theoretical levels of logging disturbance using CDA, and D) using ECA. For figure C, the dashed line corresponds to the most recent (and highest) value of CDA experienced by the river. For figure D, the dashed line corresponds to the highest value of ECA experienced by the river.

Map of MD territory watersheds

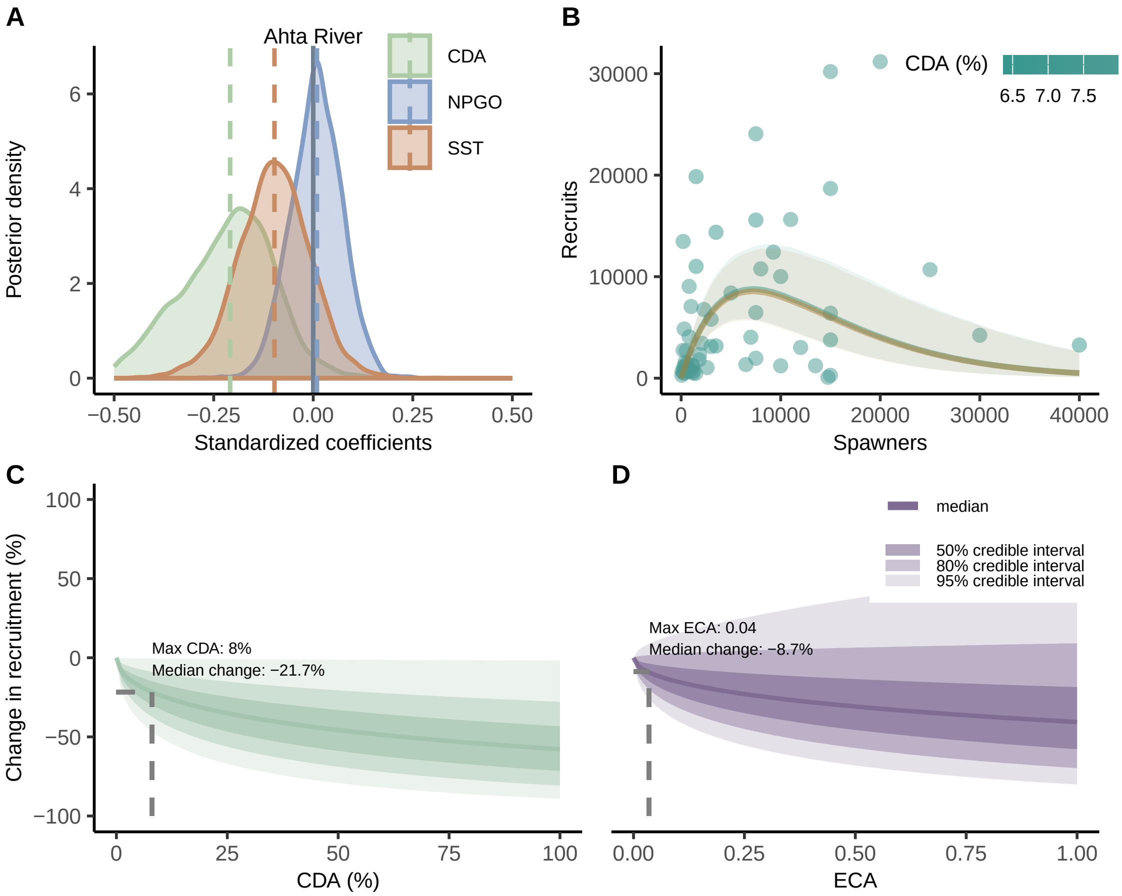


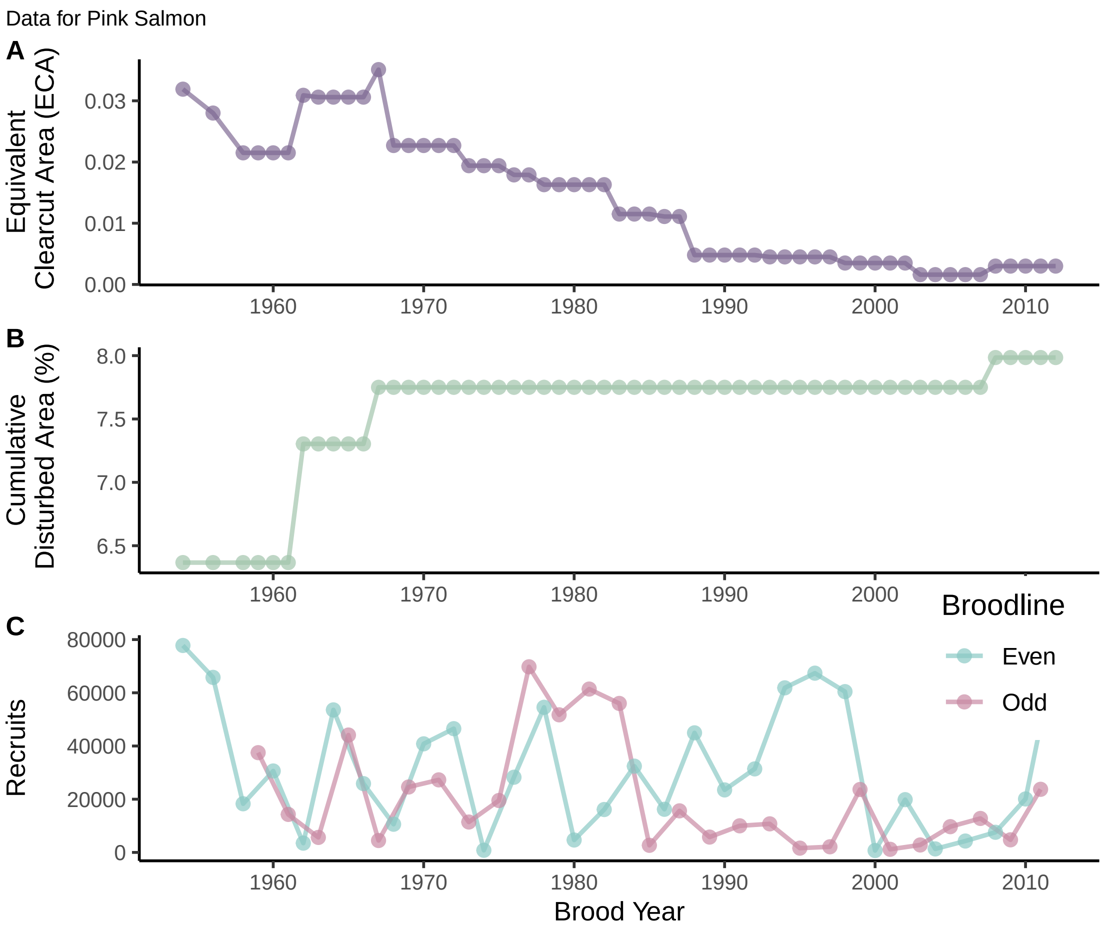
River-level data and model results

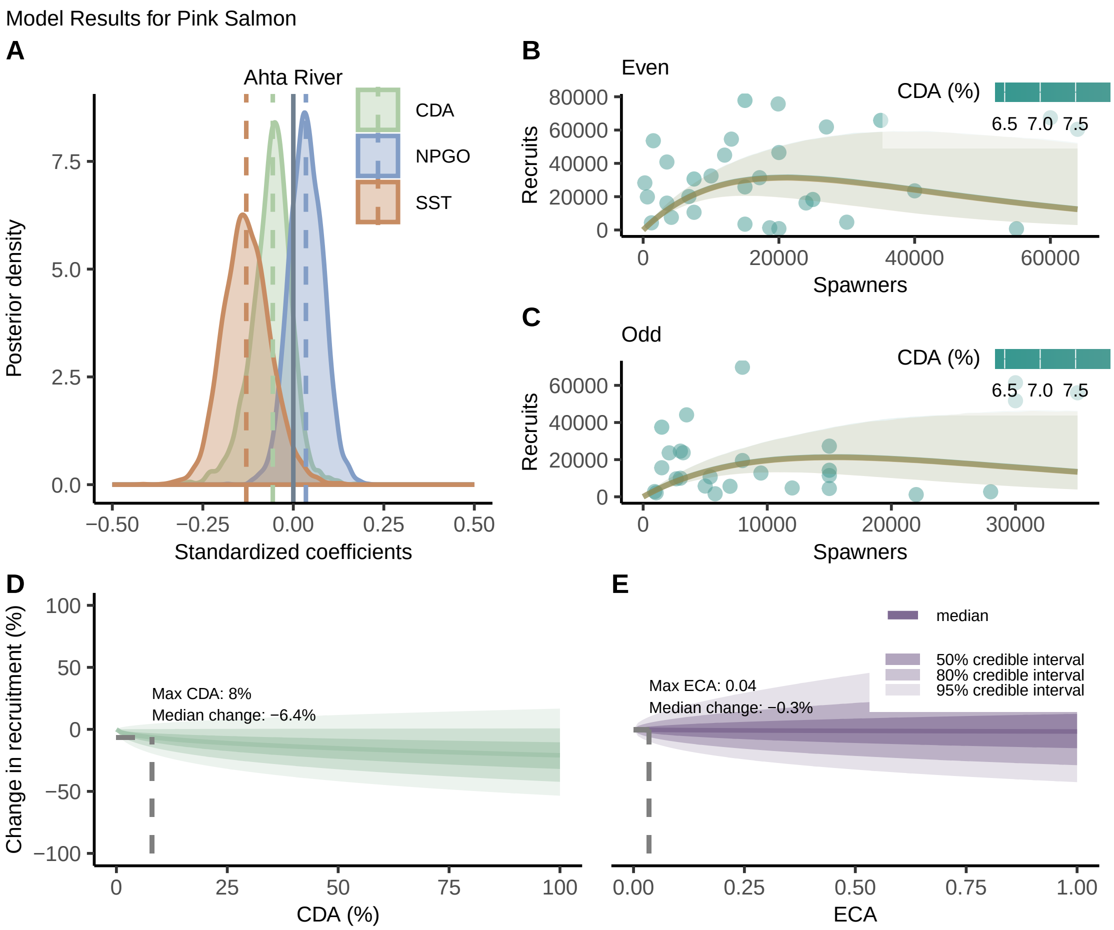
AHTA RIVER



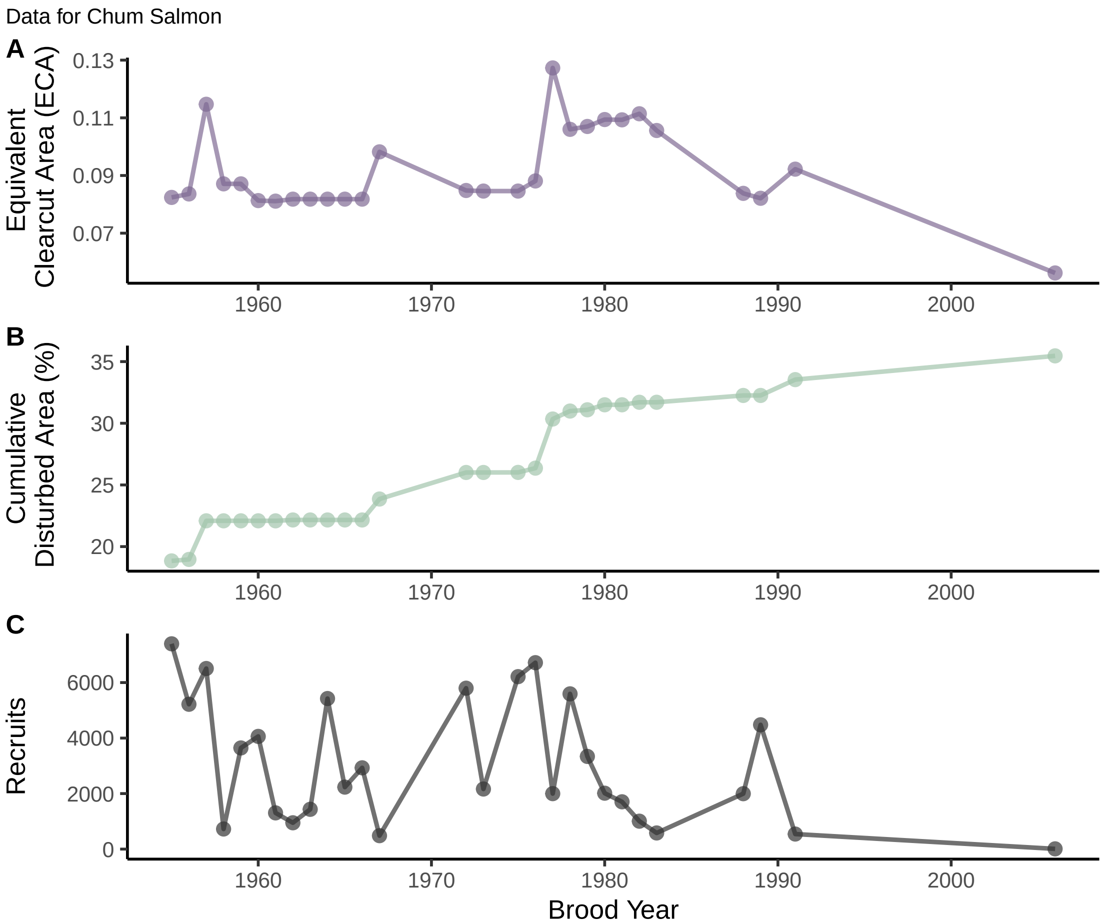
Model Results for Chum Salmon

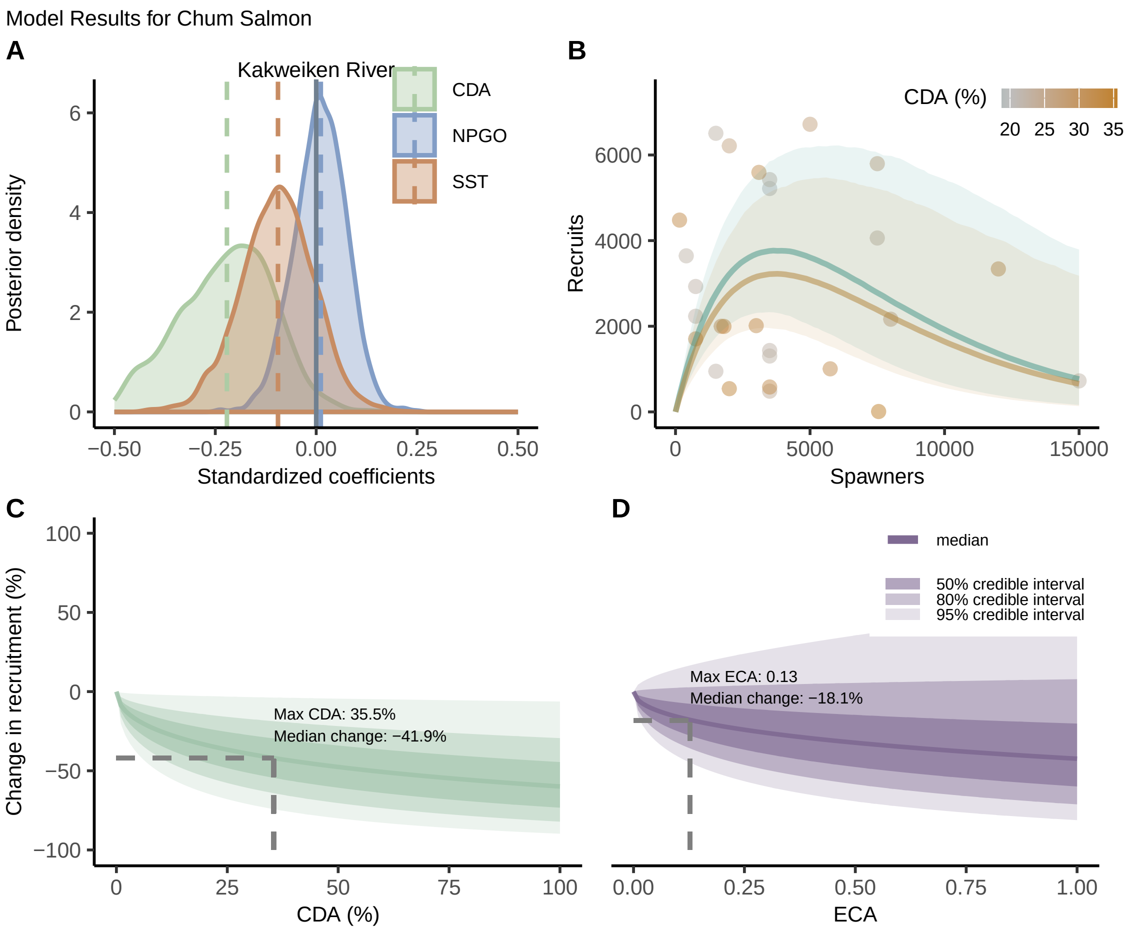



Data for Pink Salmon
A
Equivalent
Clearcut Area (ECA)
0.03
0.02
0.01
0.00
1960
1970
1980
1990
2000
2010
B
Cumulative
Disturbed Area (%)
8.0
7.5
7.0
6.5
C
Recruits
80000
60000
40000
20000
0
Broodline
Even
Odd
Brood Year


Model Results for Pink Salmon
A
Ahta River
CDA
NPGO
SST
Posterior density
Standardized coefficients
B
Even
CDA (%)
6.5 7.0 7.5
Recruits
Spawners
C
Odd
CDA (%)
6.5 7.0 7.5
Recruits
Spawners
D
Max CDA: 8%
Median change: −6.4%
Change in recruitment (%)
CDA (%)
E
median
50% credible interval
80% credible interval
95% credible interval
Max ECA: 0.04
Median change: −0.3%
ECA

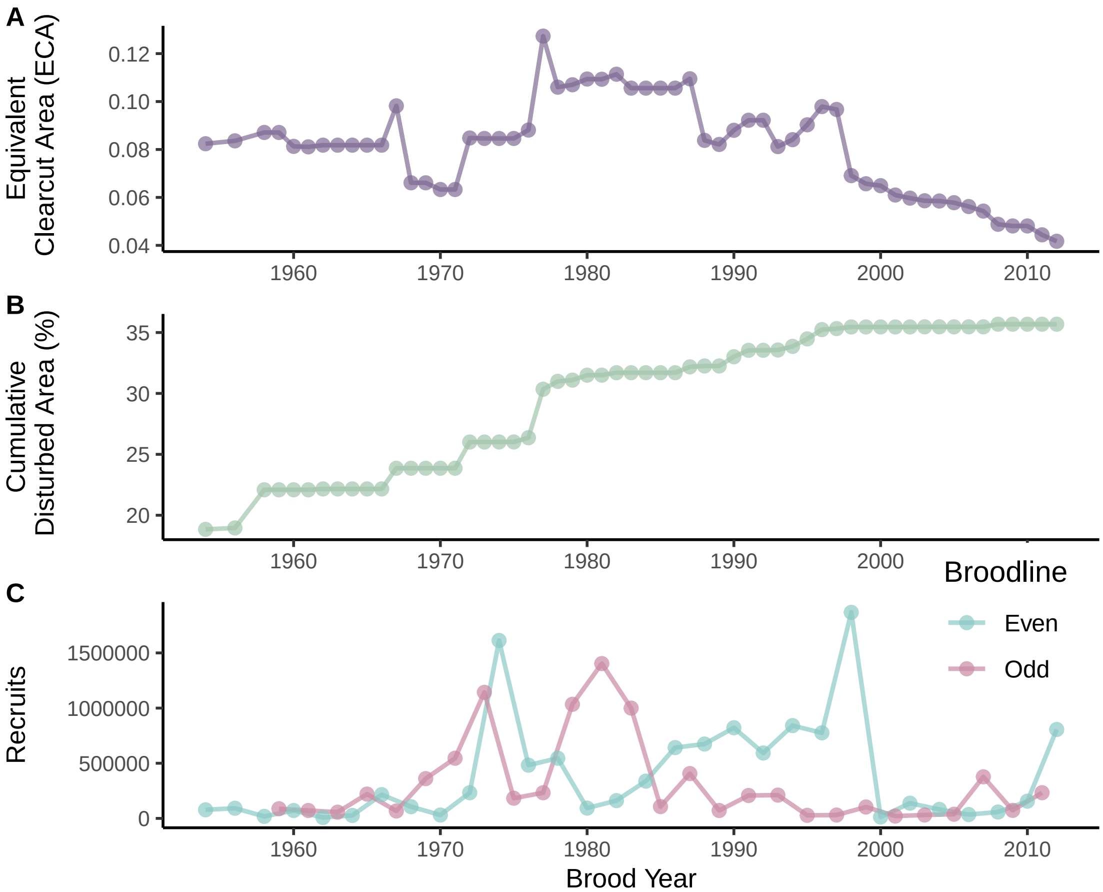
KAKWEIKEN RIVER

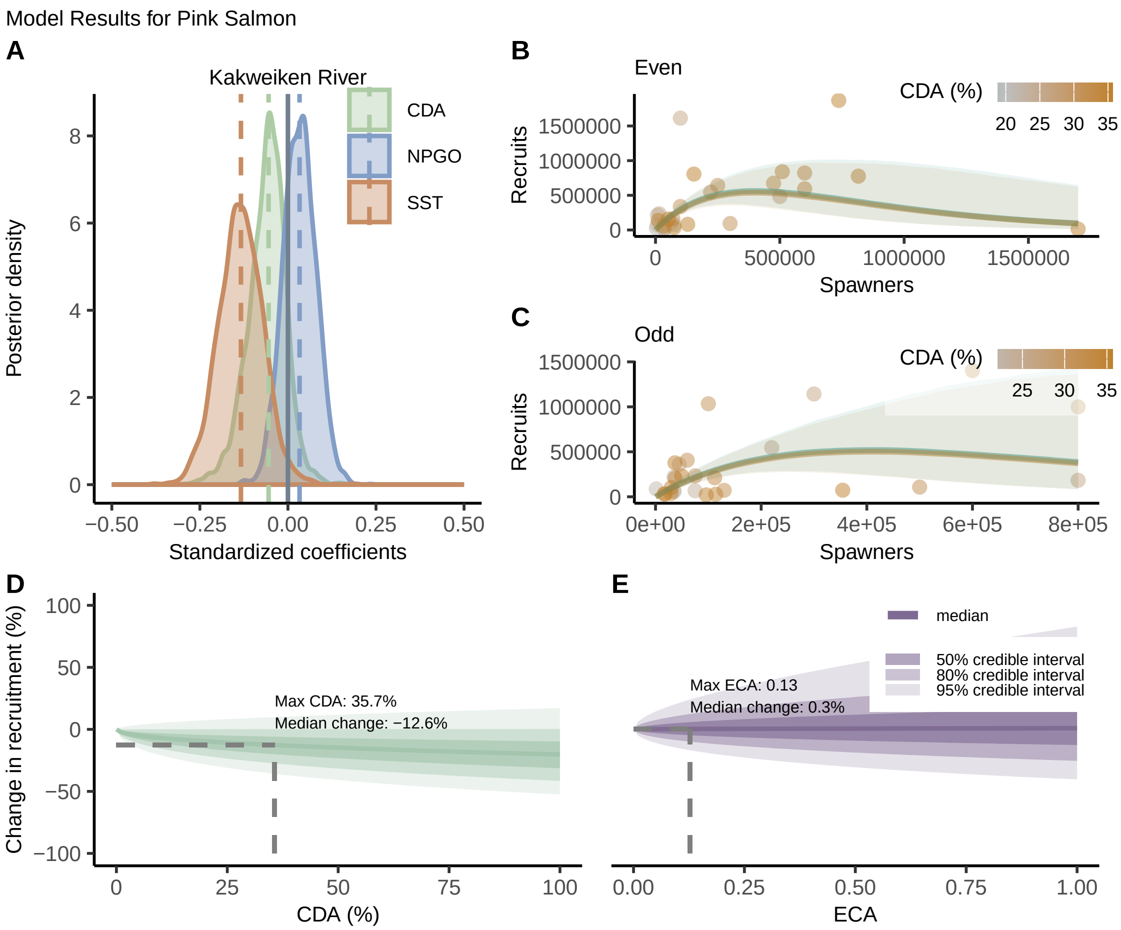




Model Results for Chum Salmon
A
Kakweiken River
CDA
NPGO
SST
Posterior density
Standardized coefficients
B
CDA (%)
20 25 30 35
Recruits
Spawners
C
Max CDA: 35.5%
Median change: −41.9%
Change in recruitment (%)
CDA (%)
D
median
50% credible interval
80% credible interval
95% credible interval
Max ECA: 0.13
Median change: −18.1%
ECA

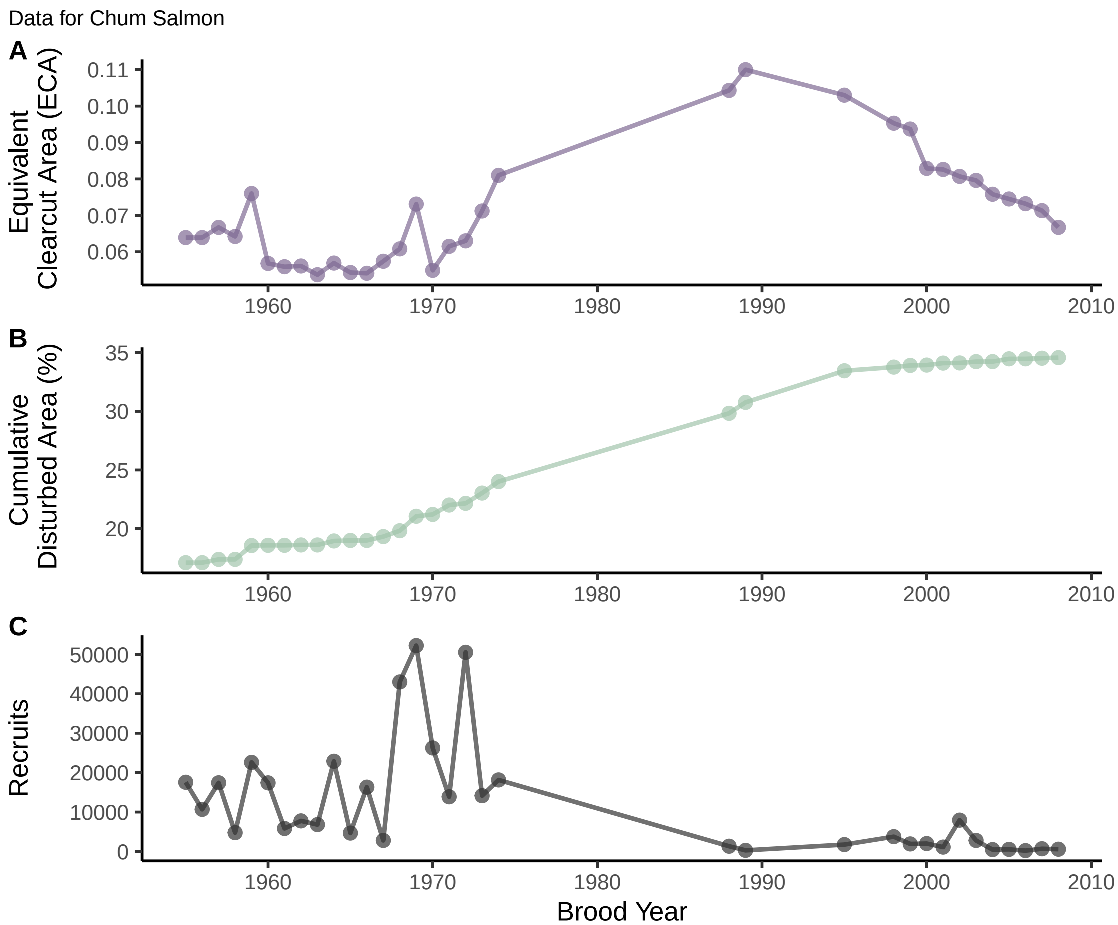
Data for Pink Salmon



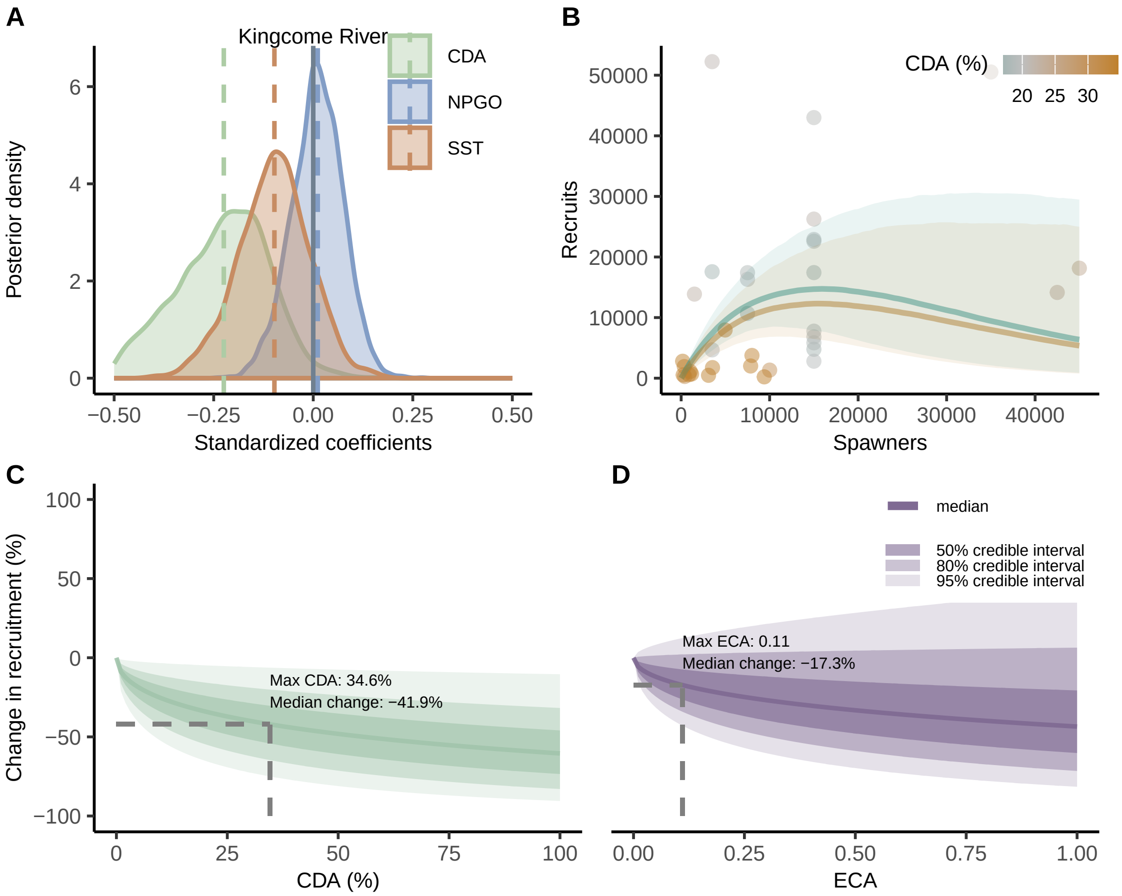


Model Results for Pink Salmon
A
Kakweiken River
CDA
NPGO
SST
Posterior density
Standardized coefficients
B
Even
CDA (%)
Recruits
Spawners
C
Odd
D
Max CDA: 35.7%
Median change: −12.6%
Change in recruitment (%)
CDA (%)
E
median
50% credible interval
80% credible interval
95% credible interval
Max ECA: 0.13
Median change: 0.3%
ECA

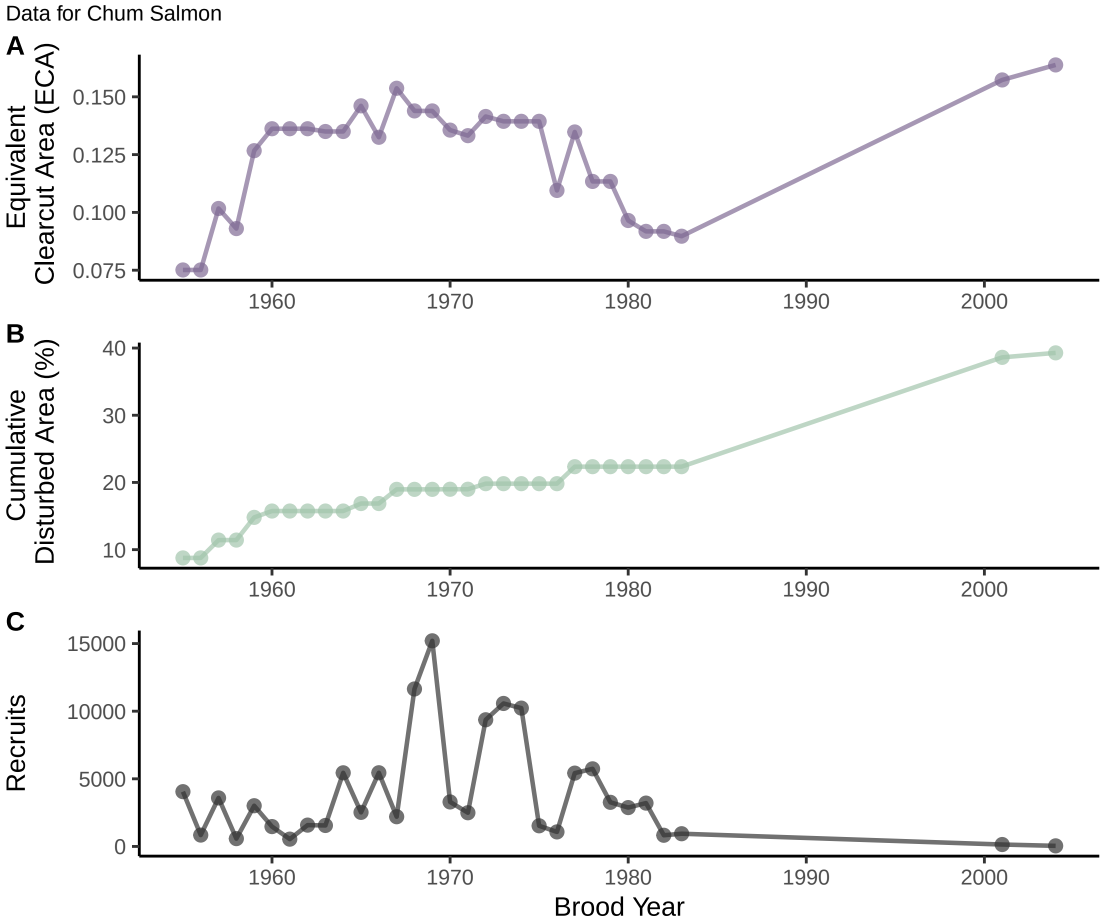
KINGCOME RIVER



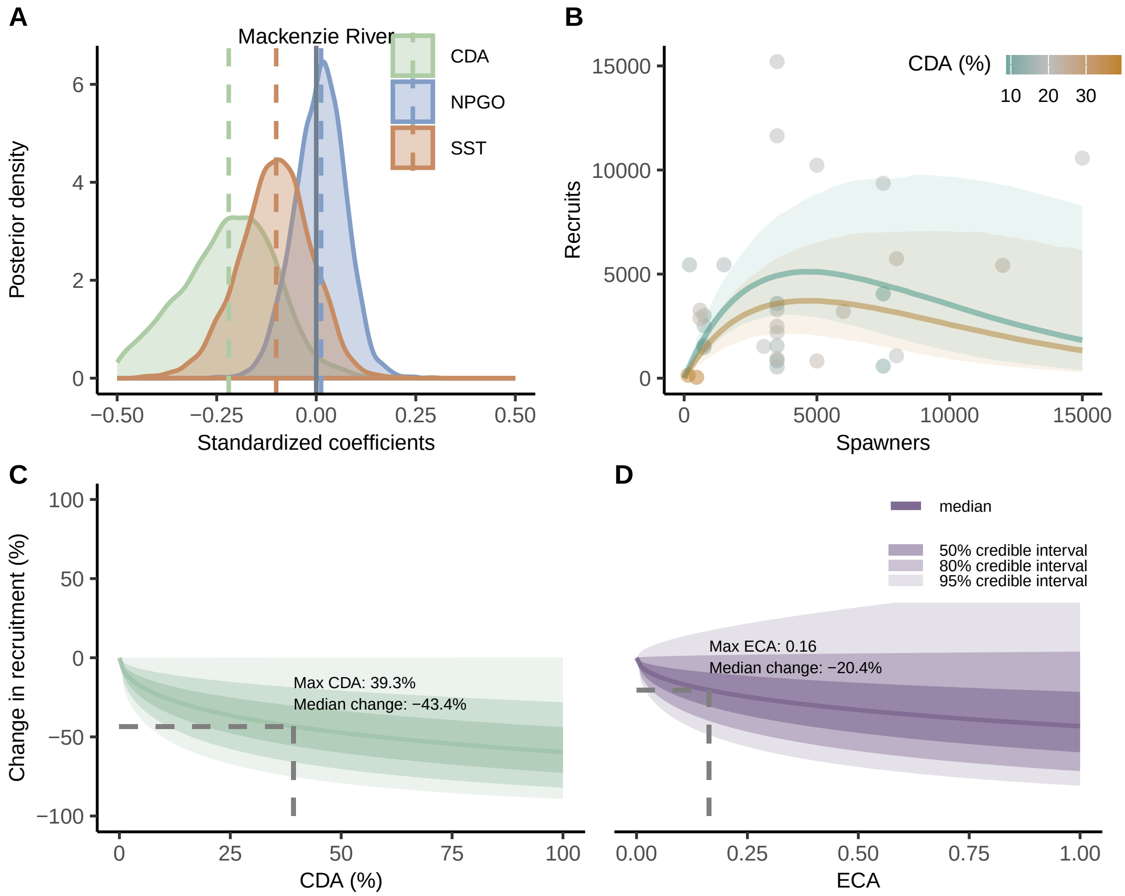
Model Results for Chum Salmon



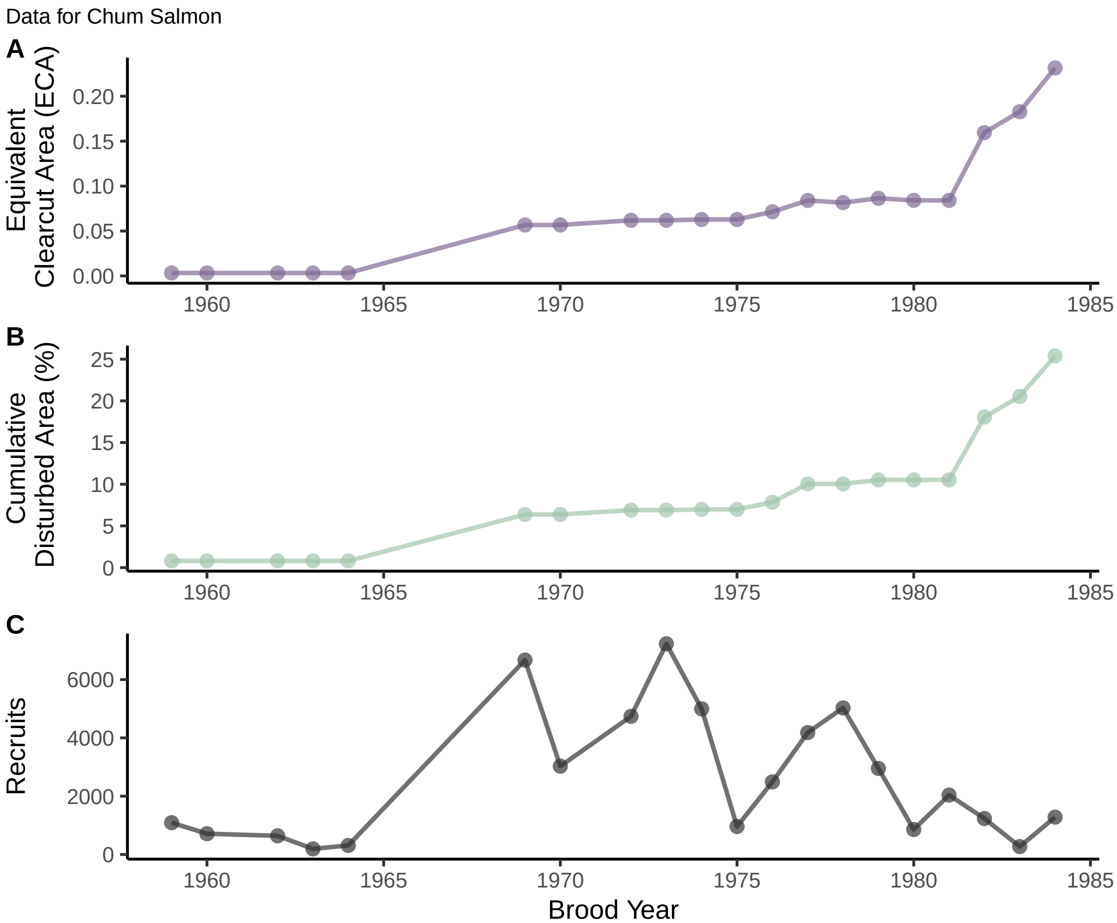
MACKENZIE RIVER



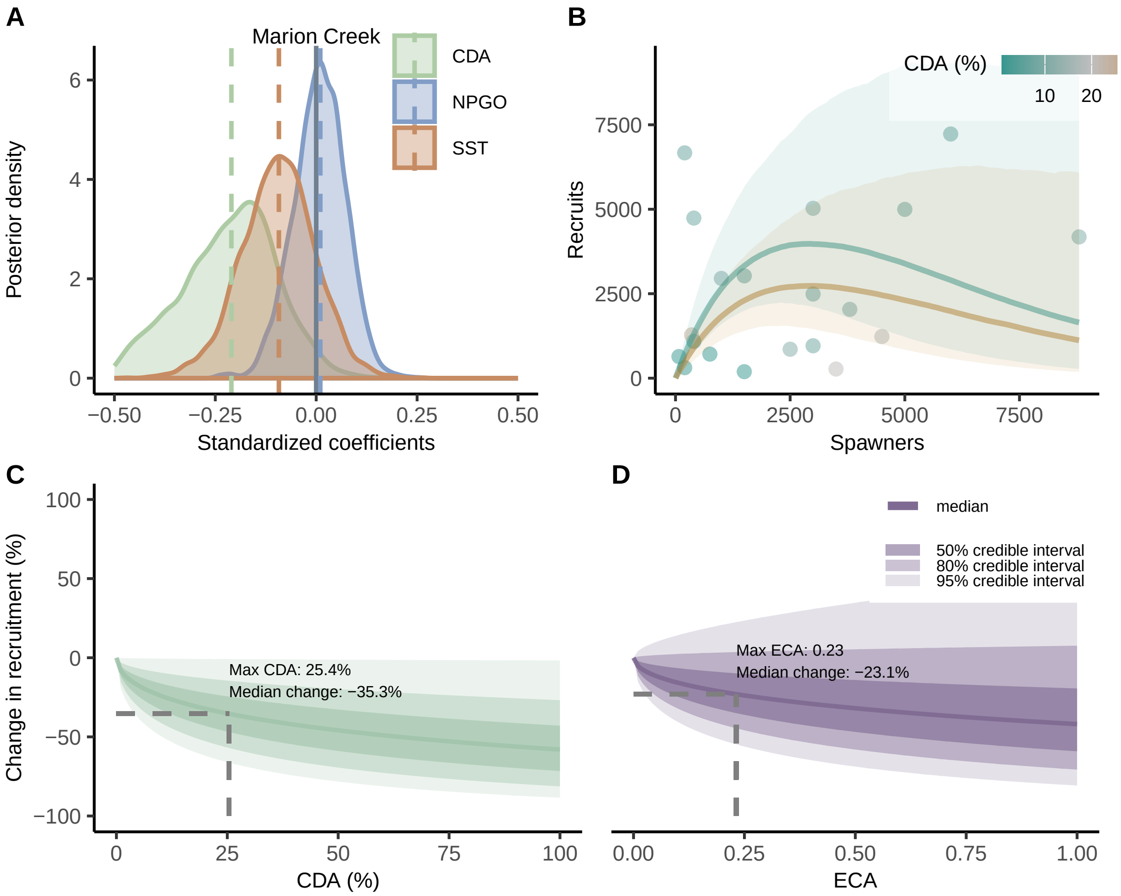
Model Results for Chum Salmon



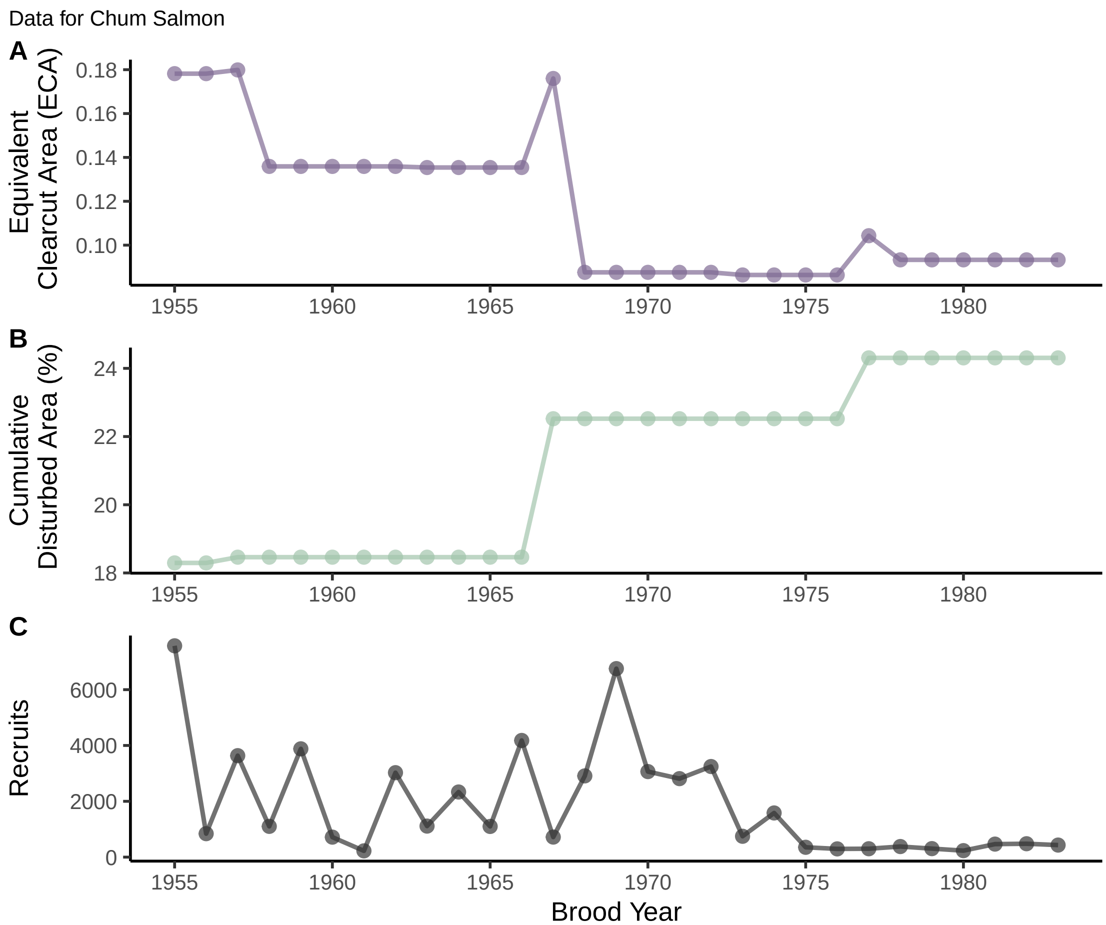
MARION CREEK



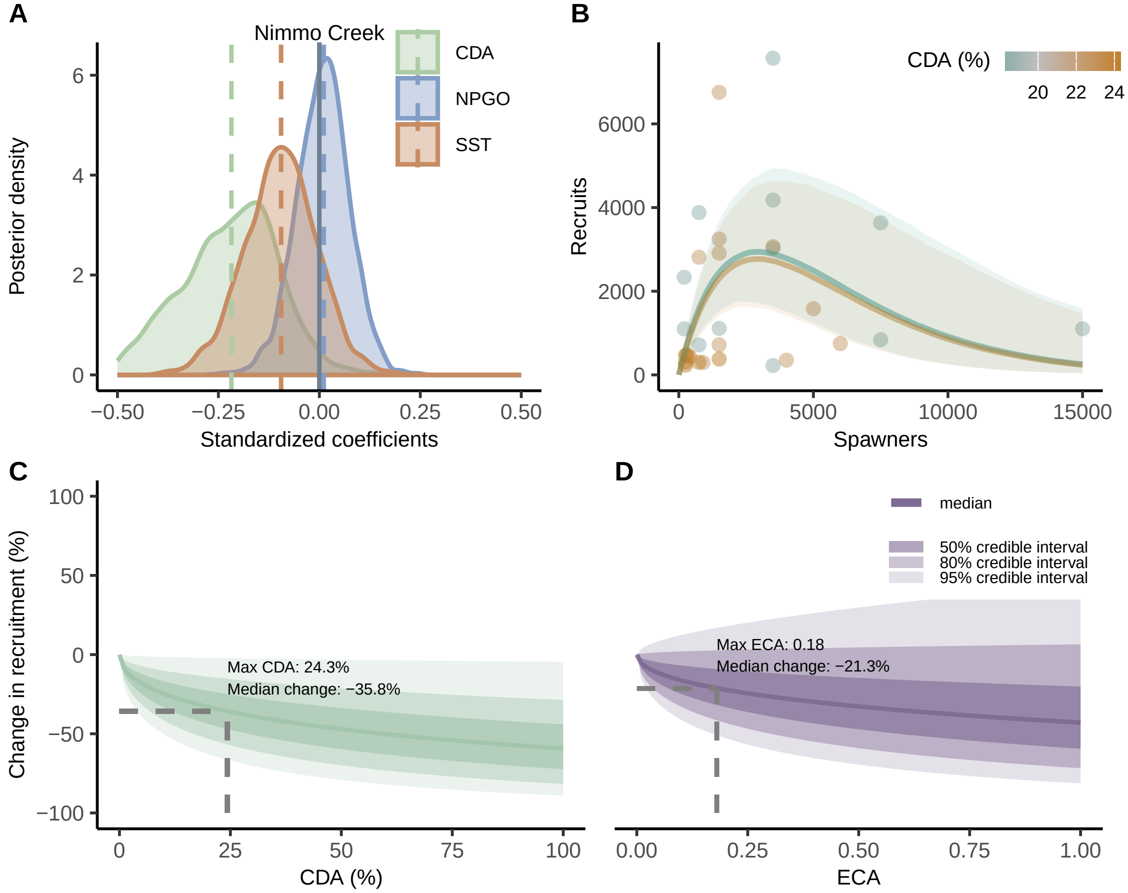
Model Results for Chum Salmon



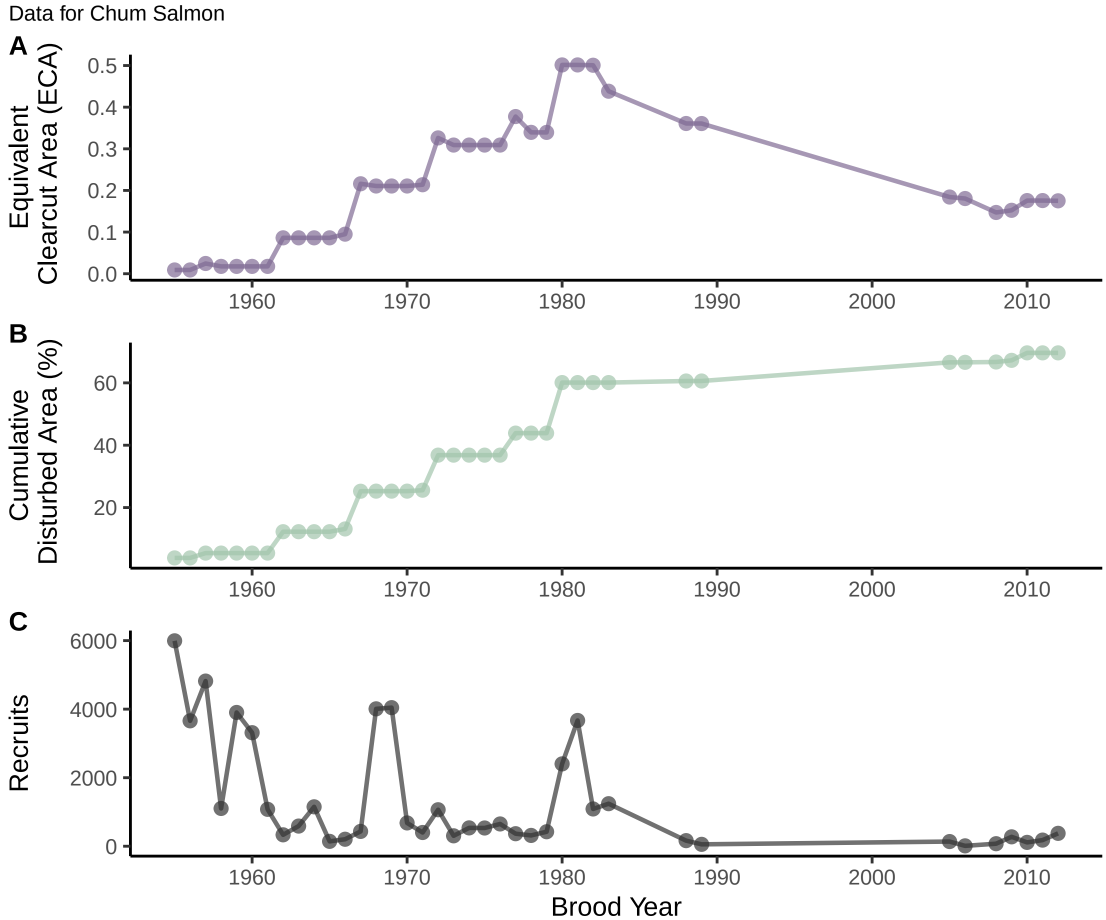
NIMMO CREEK


Data for Chum Salmon
A
Equivalent Clearcut Area (ECA)
0.18
0.16
0.14
0.12
0.10
1955
1960
1965
1970
1975
1980
B
Cumulative Disturbed Area (%)
24
22
20
18
1955
1960
1965
1970
1975
1980
C
Recruits
6000
4000
2000
0
1955
1960
1965
1970
1975
1980
Brood Year

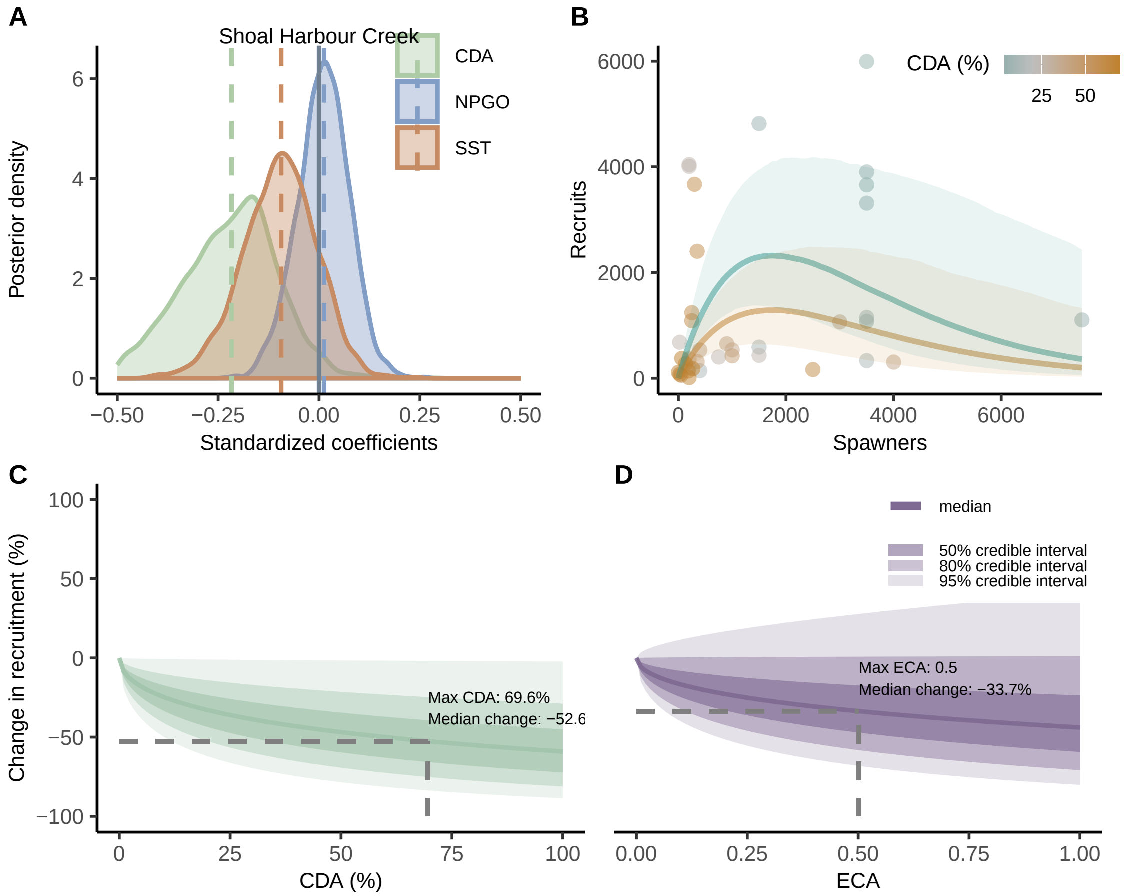
Model Results for Chum Salmon



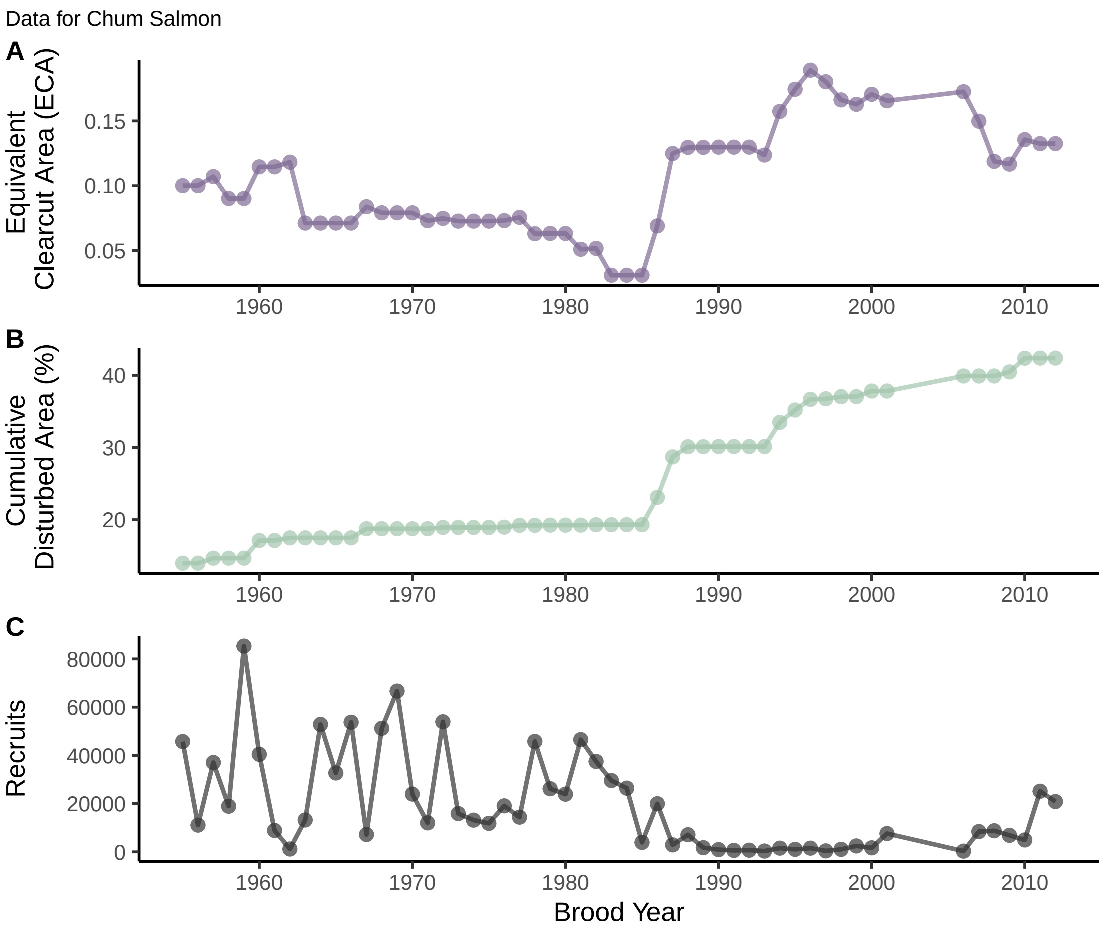
SHOAL HARBOUR CREEK

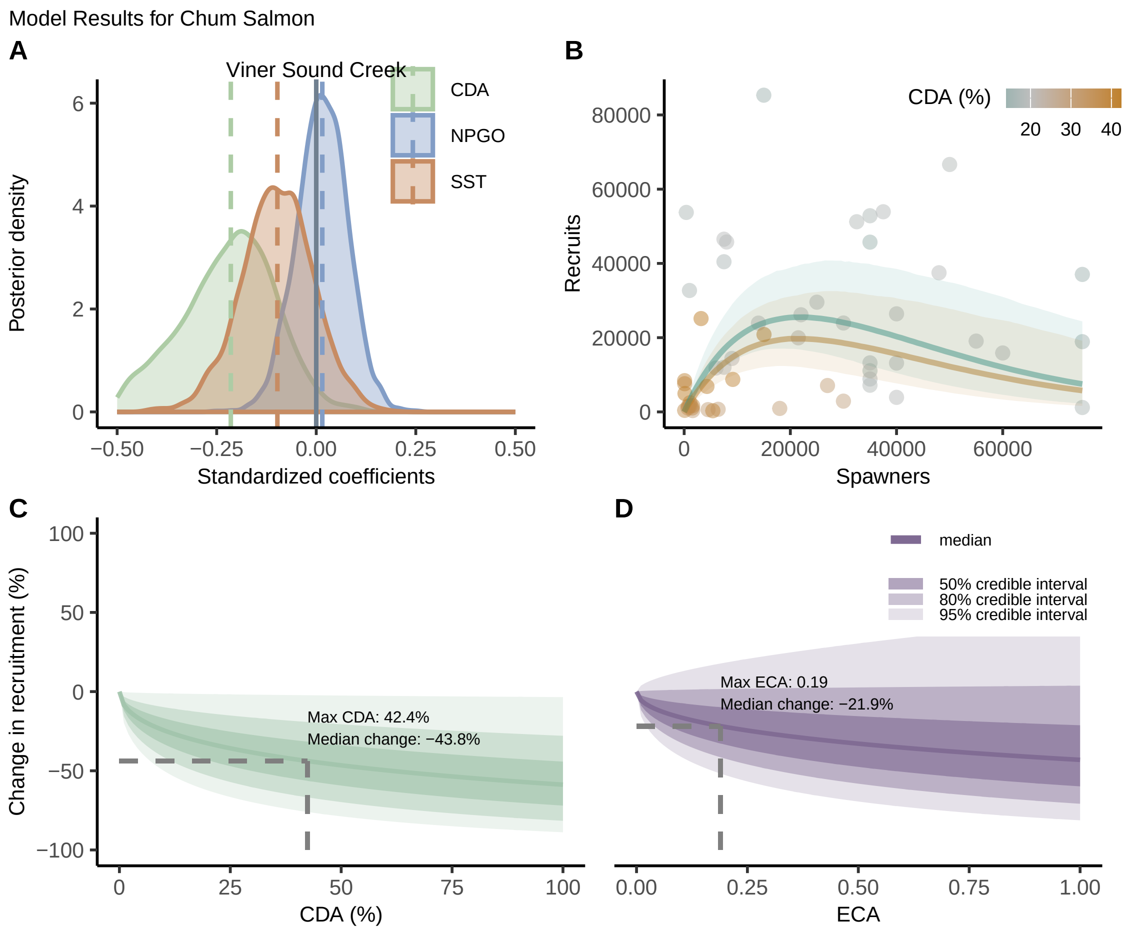


Model Results for Chum Salmon



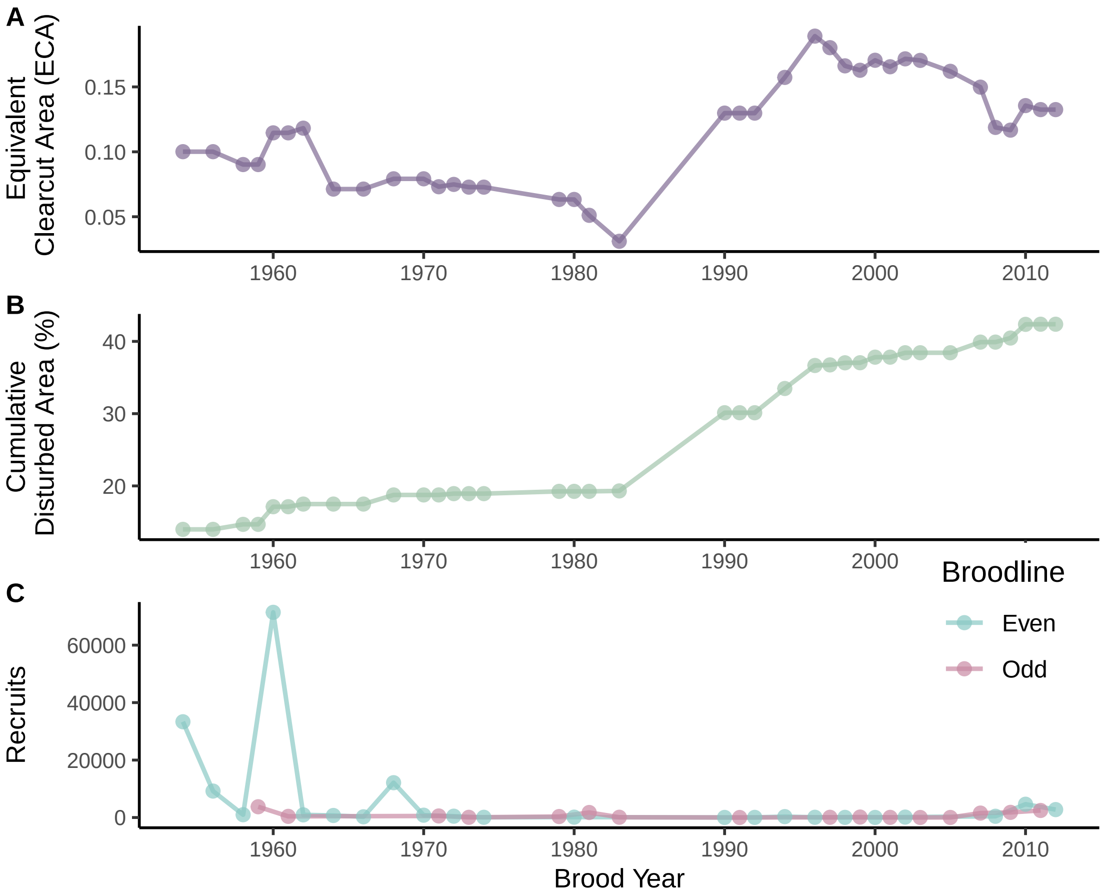
VINER SOUND CREEK

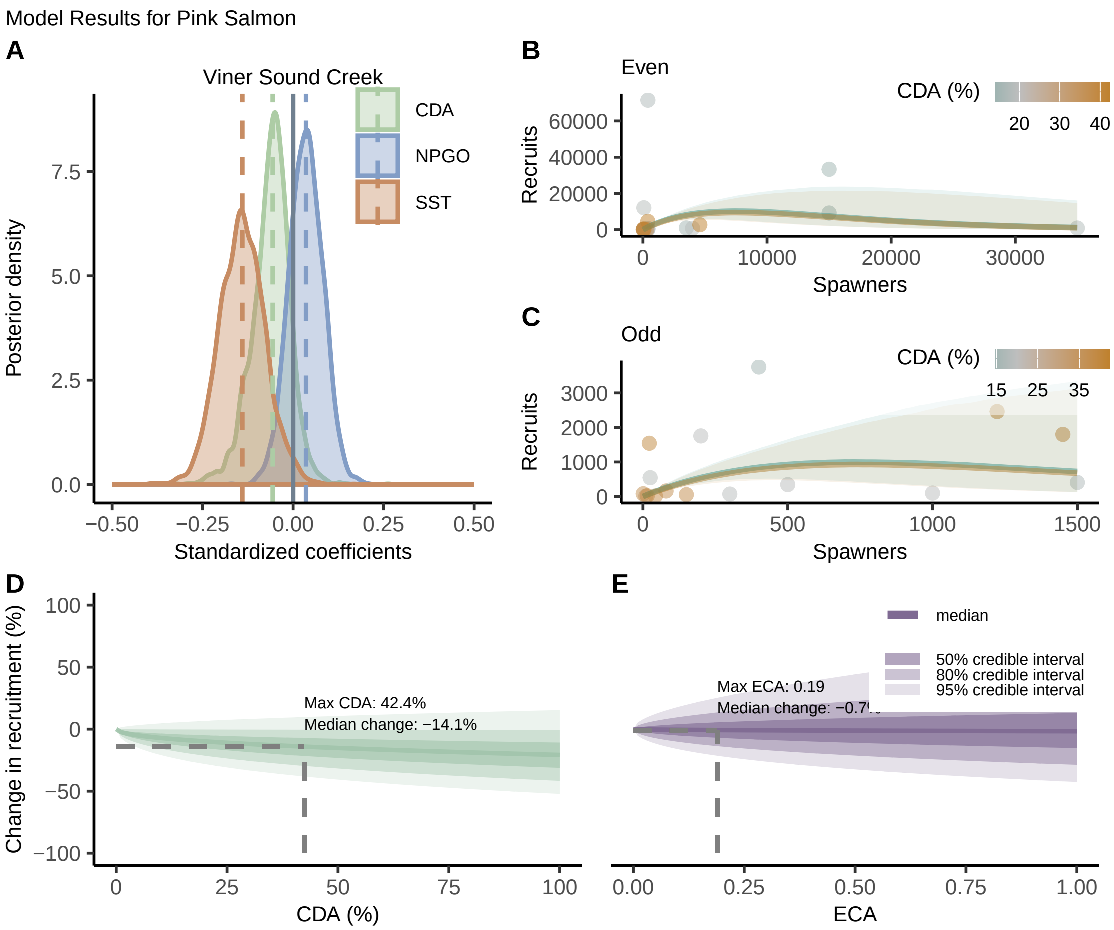




Model Results for Chum Salmon
A
Viner Sound Creek
CDA
NPGO
SST
Posterior density
Standardized coefficients
B
CDA (%)
20 30 40
Recruits
Spawners
C
Change in recruitment (%)
Max CDA: 42.4%
Median change: −43.8%
CDA (%)
D
median
50% credible interval
80% credible interval
95% credible interval
Max ECA: 0.19
Median change: −21.9%
ECA

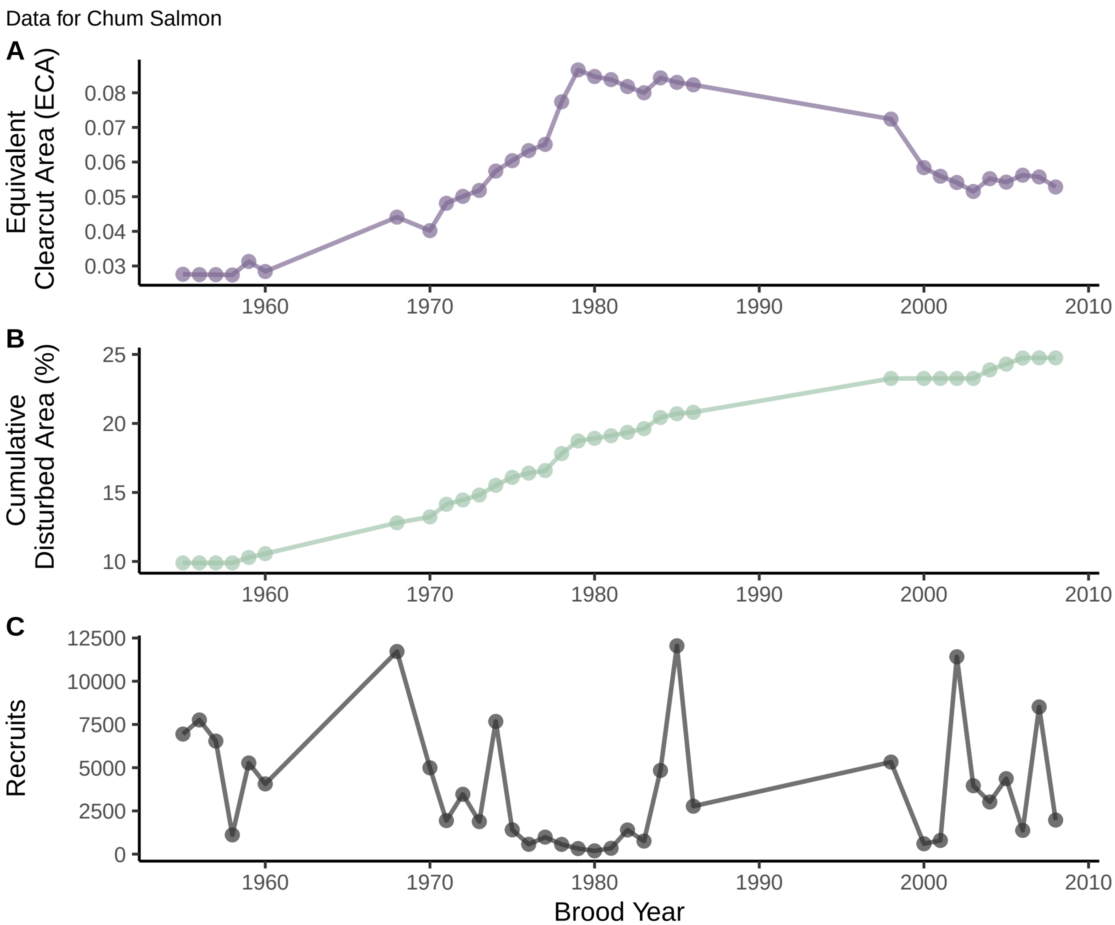
Data for Pink Salmon



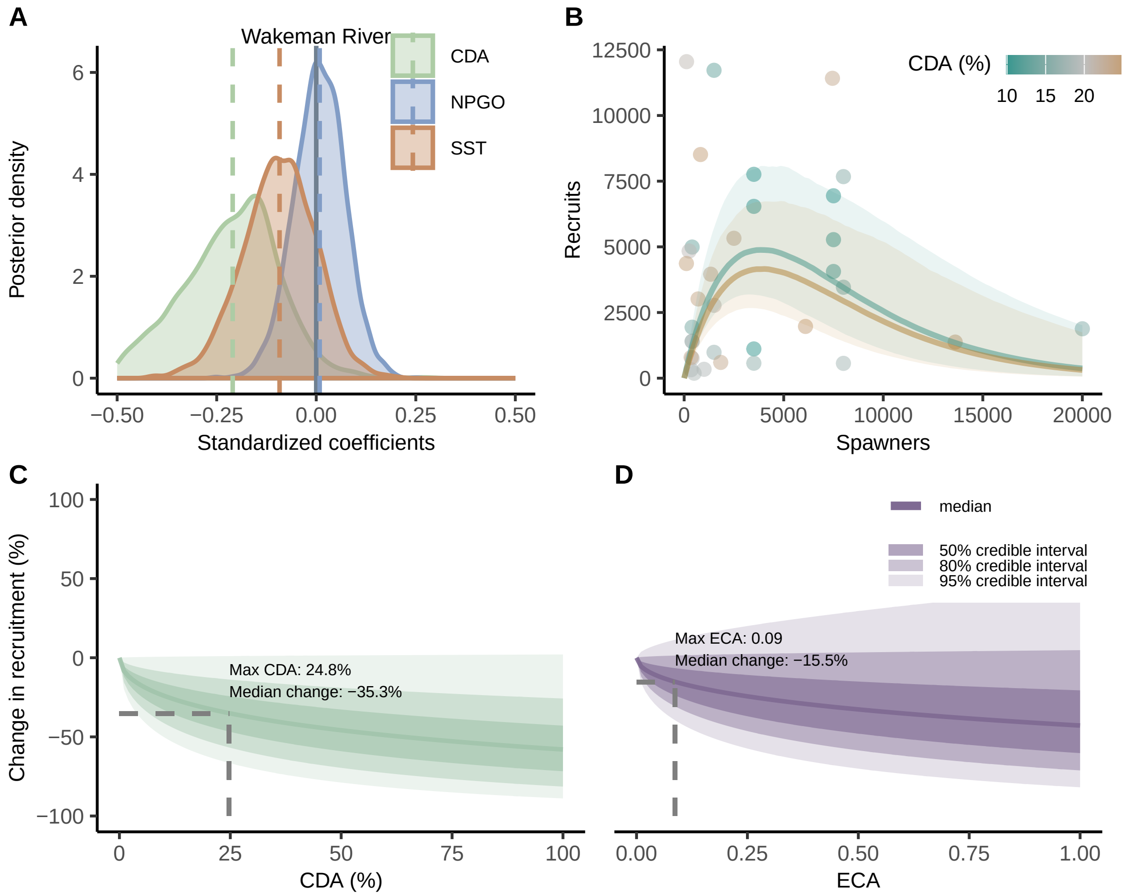


Model Results for Pink Salmon
A
Viner Sound Creek
CDA
NPGO
SST
Posterior density
Standardized coefficients
B
Even
CDA (%)
20 30 40
Recruits
Spawners
C
Odd
CDA (%)
15 25 35
Recruits
Spawners
D
Max CDA: 42.4%
Median change: −14.1%
Change in recruitment (%)
CDA (%)
E
median
50% credible interval
80% credible interval
95% credible interval
Max ECA: 0.19
ECA

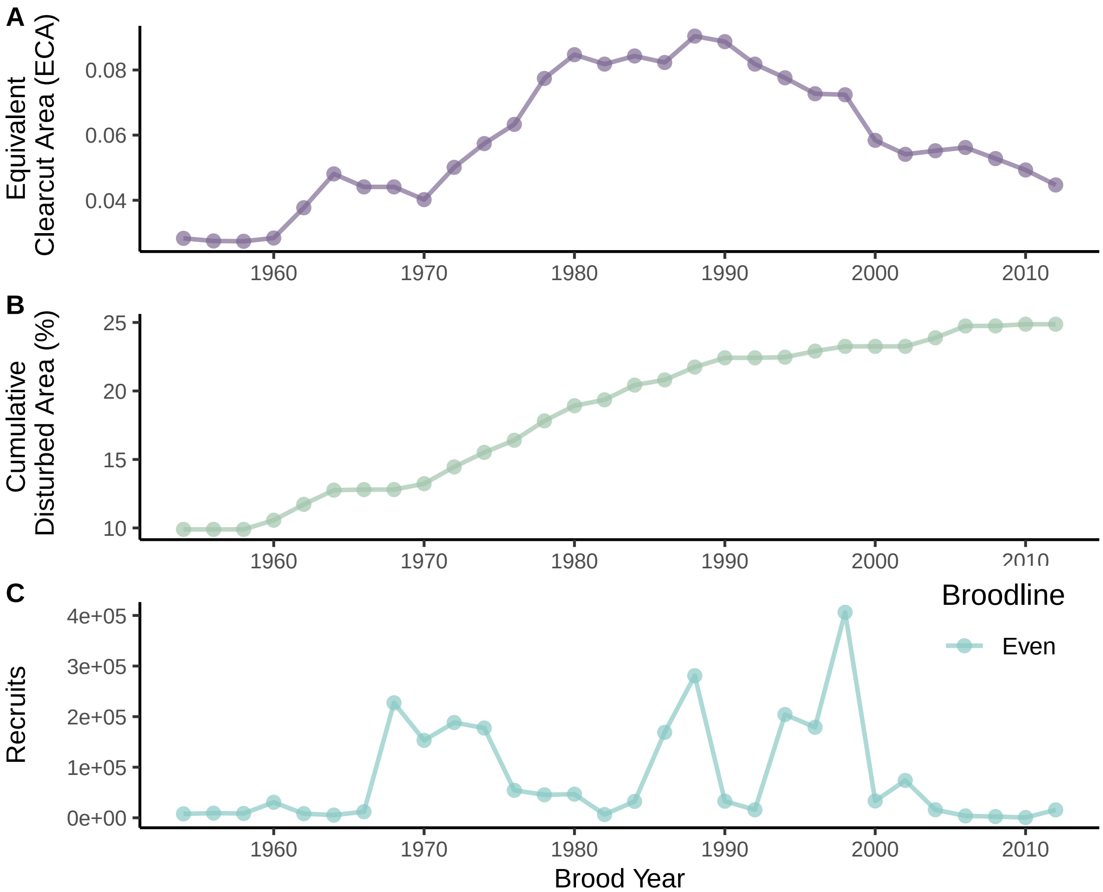
WAKEMAN RIVER



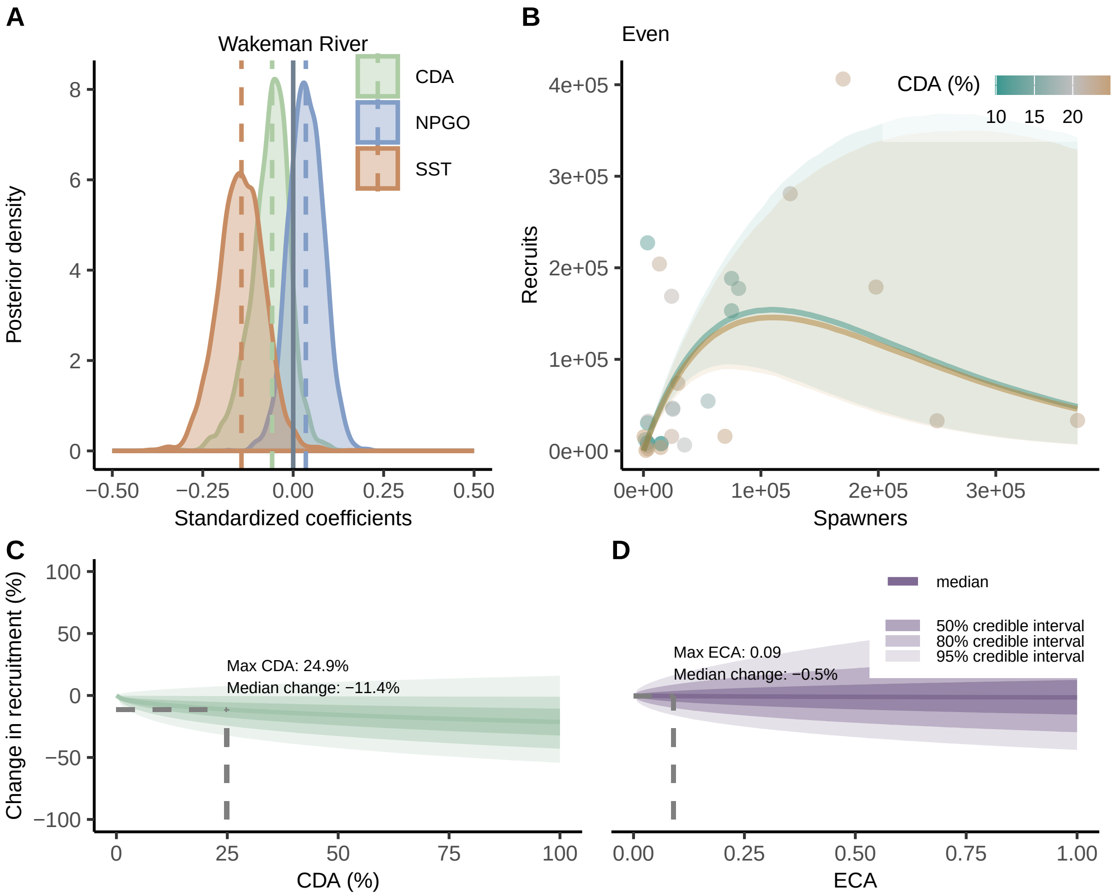
Model Results for Chum Salmon



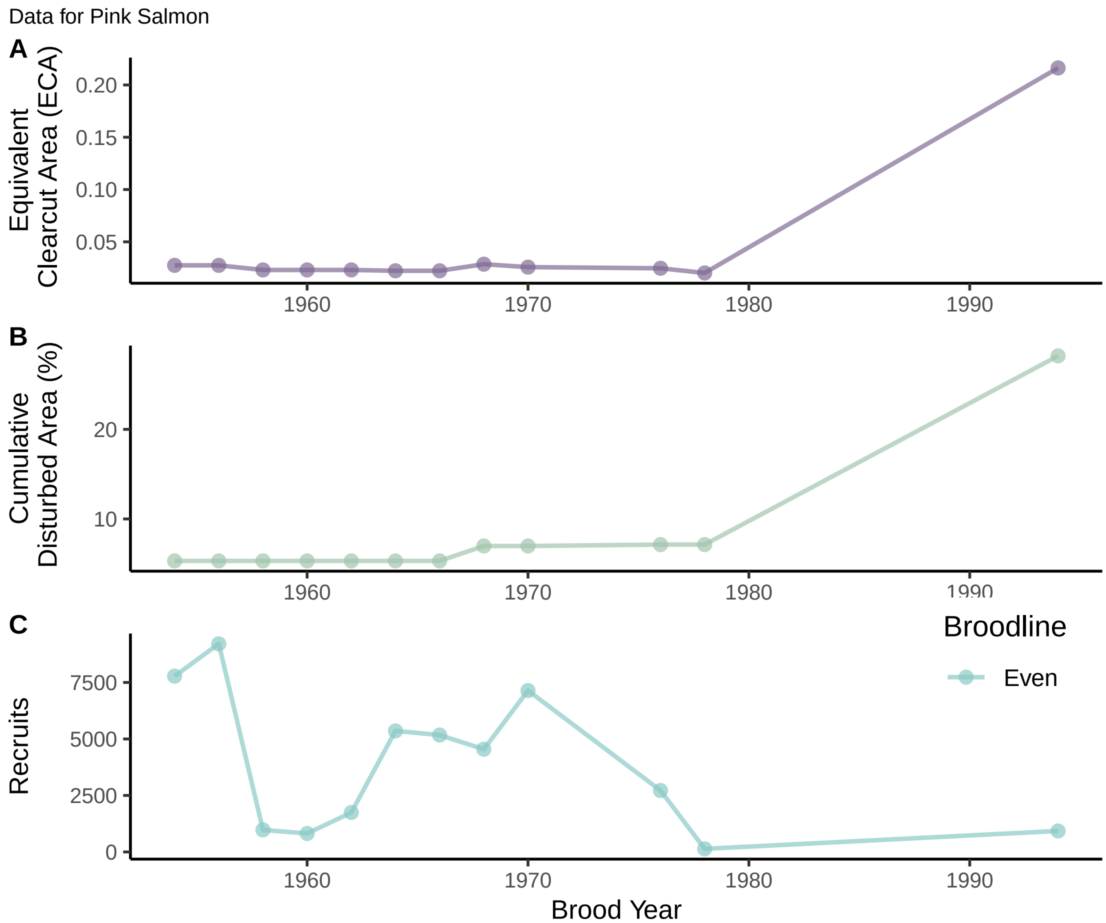
Data for Pink Salmon

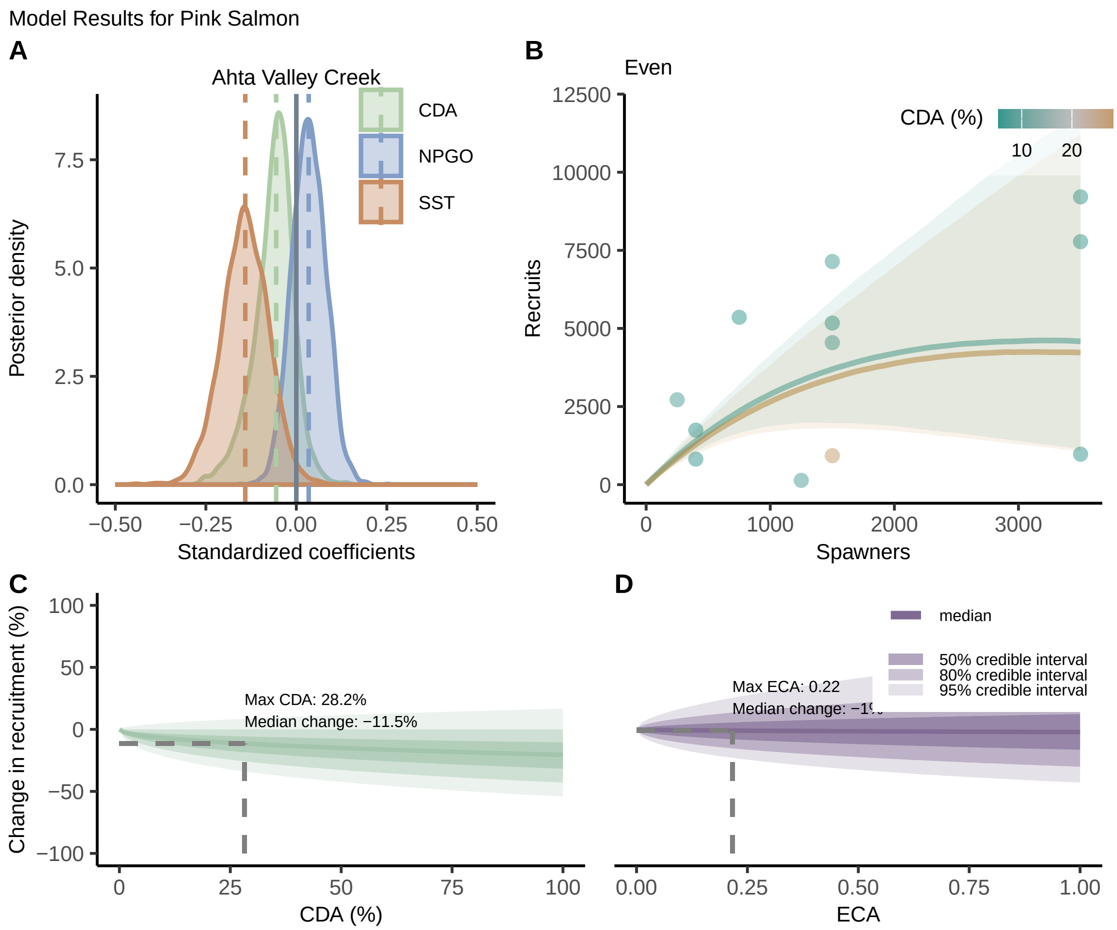


Model Results for Pink Salmon

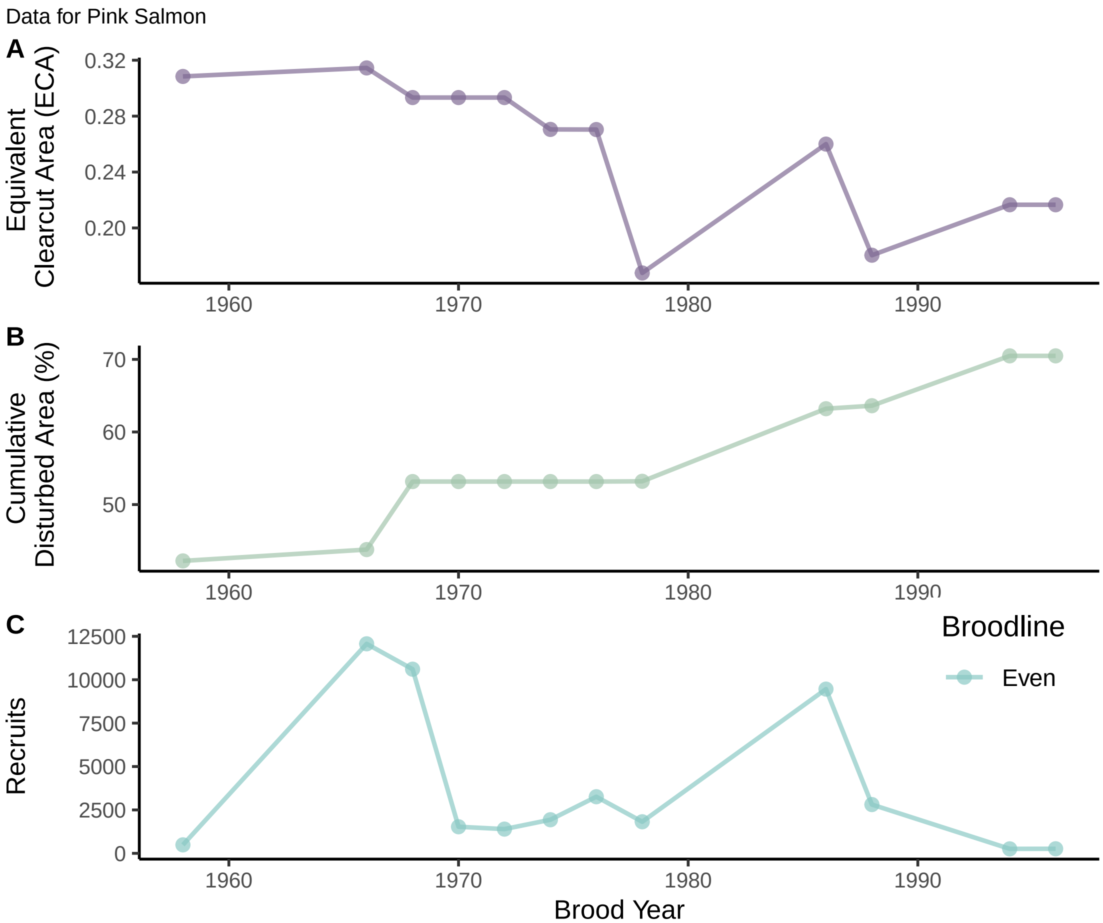


AHTA VALLEY CREEK

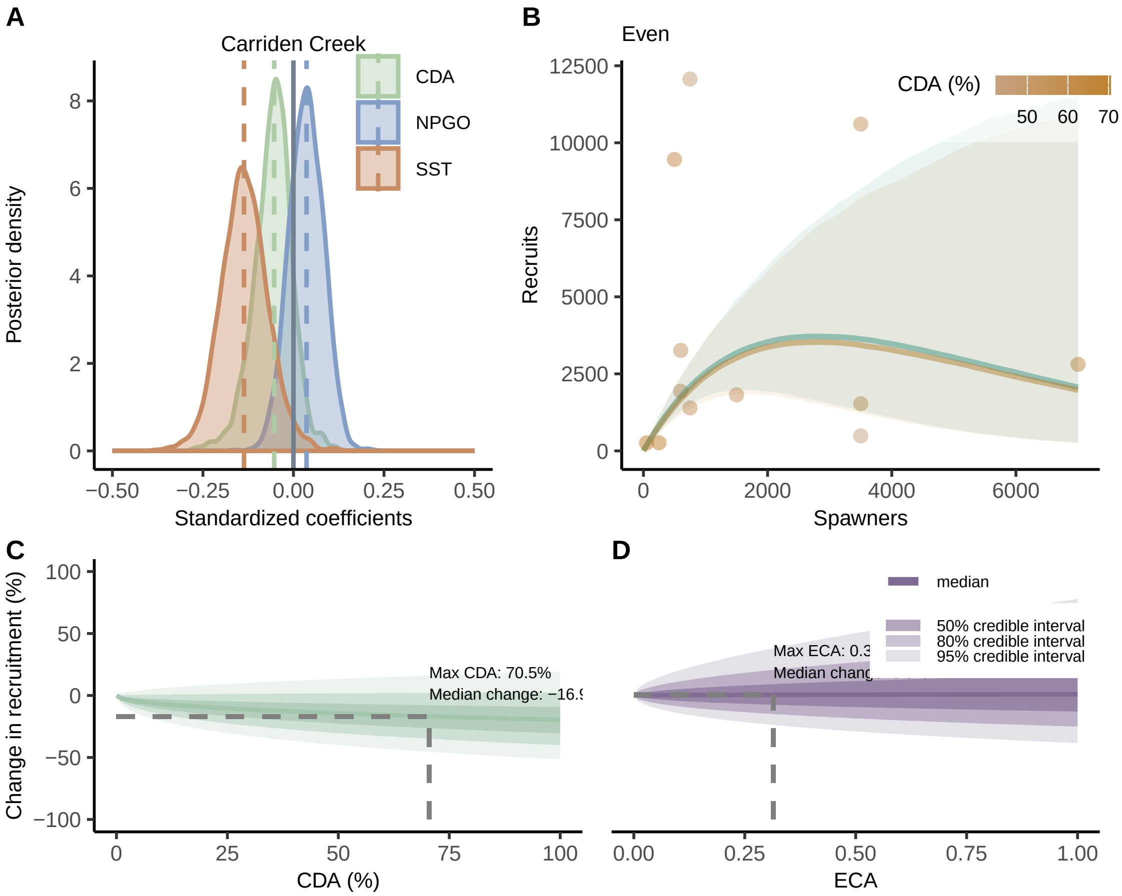



Model Results for Pink Salmon
A
Ahta Valley Creek
CDA
NPGO
SST
Posterior density
Standardized coefficients
B
Even
CDA (%)
10
20
Recruits
Spawners
C
Change in recruitment (%)
Max CDA: 28.2%
Median change: −11.5%
CDA (%)
D
median
50% credible interval
80% credible interval
95% credible interval
Max ECA: 0.22
ECA

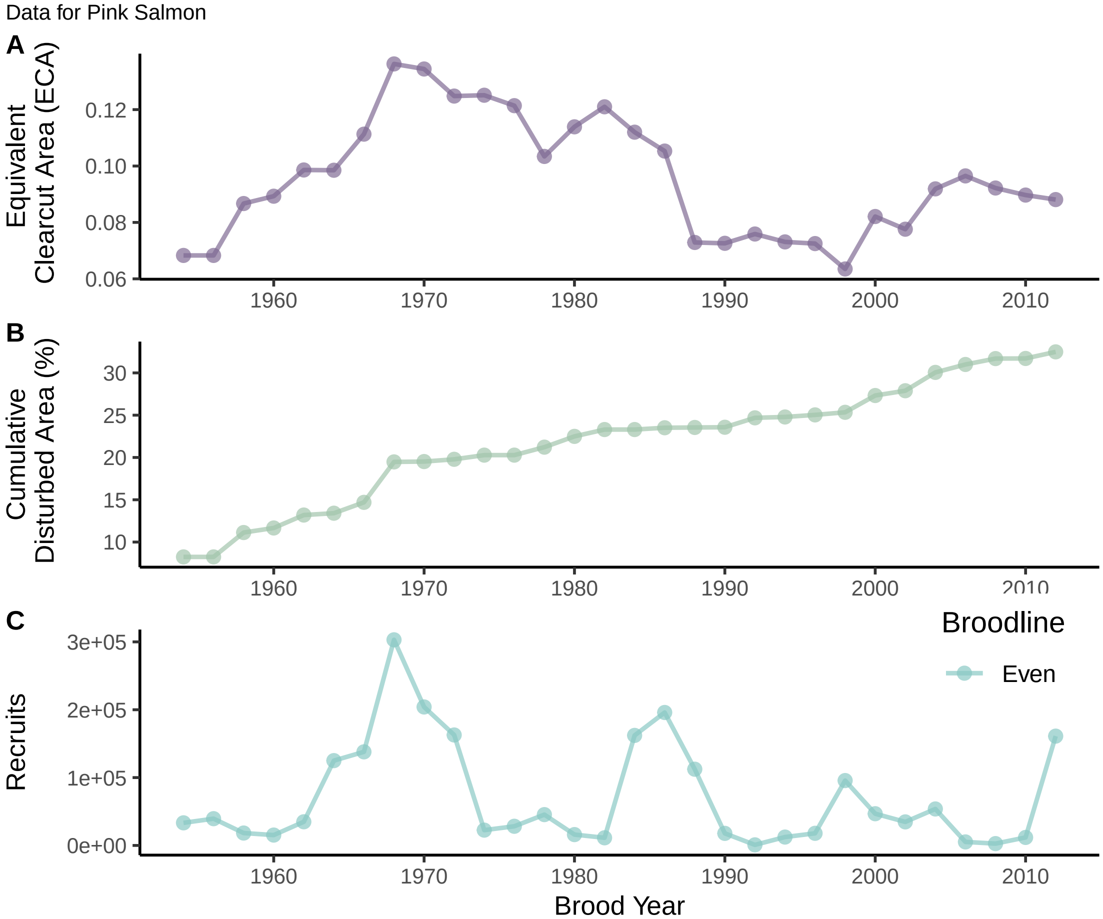
CARRIDEN CREEK

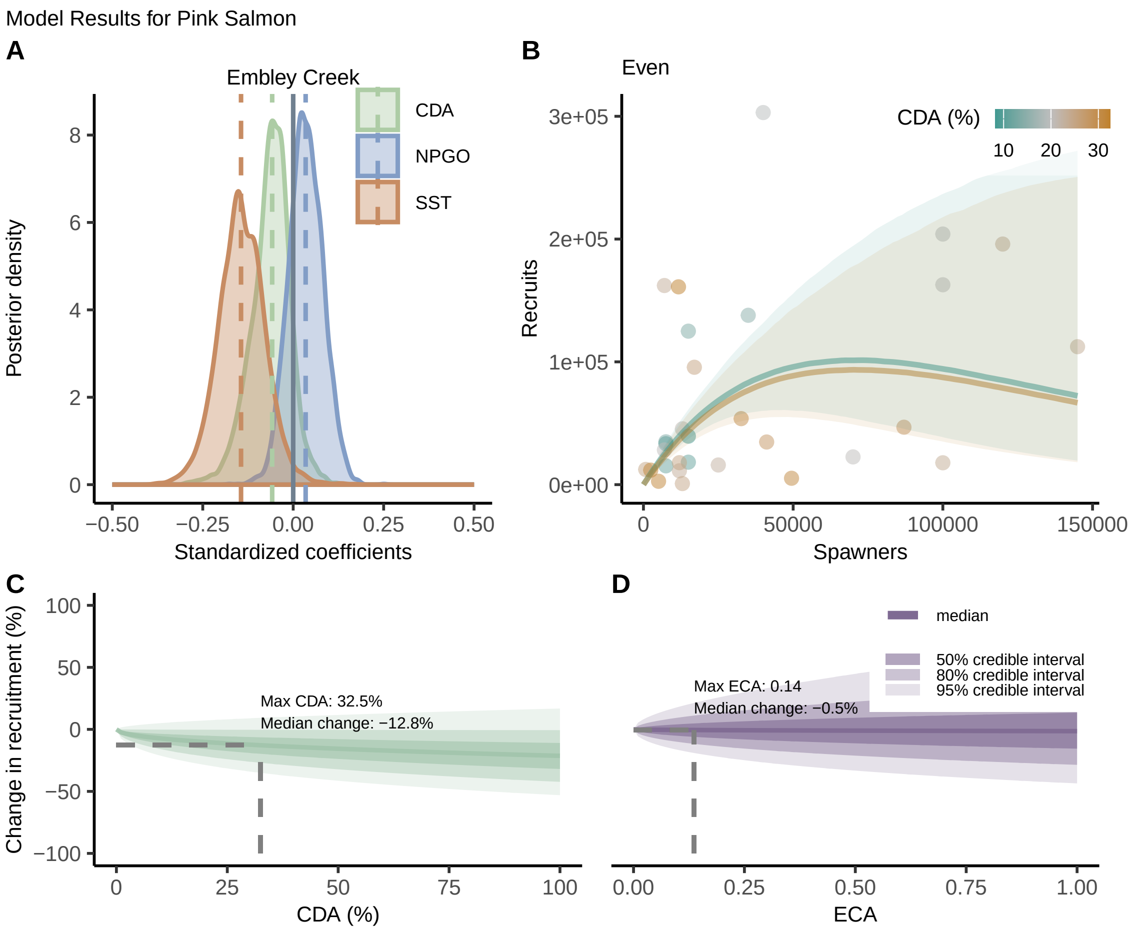


Model Results for Pink Salmon

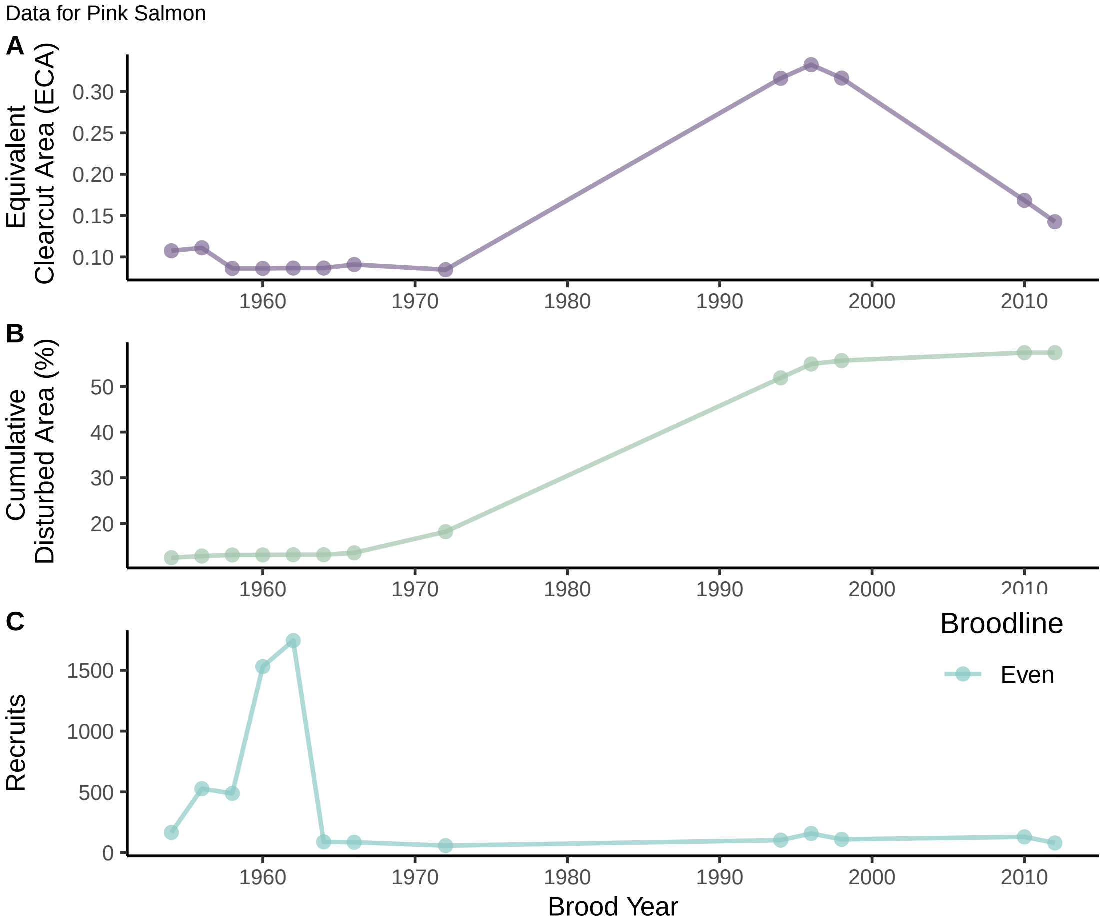


EMBLEY CREEK

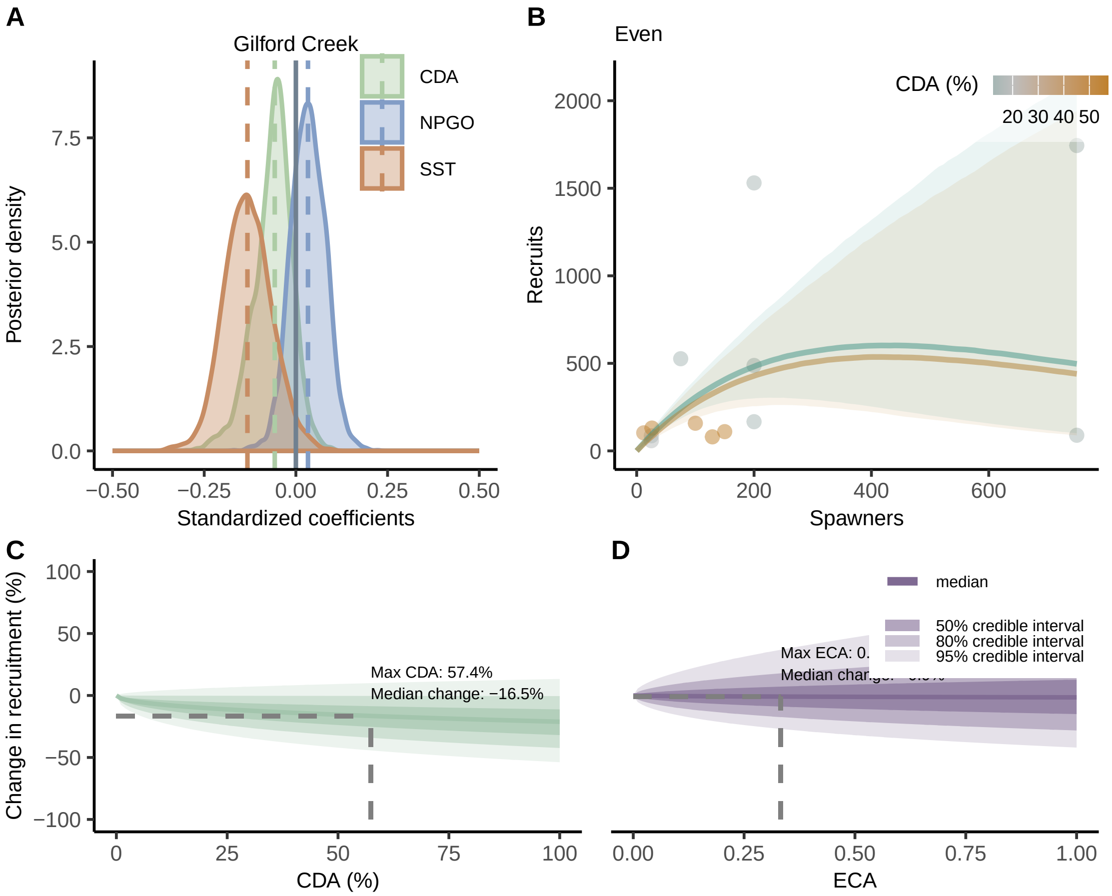



Model Results for Pink Salmon
A
Embley Creek
CDA
NPGO
SST
Posterior density
Standardized coefficients
B
Even
CDA (%)
10
20
30
Recruits
Spawners
C
Max CDA: 32.5%
Median change: −12.8%
Change in recruitment (%)
CDA (%)
D
median
50% credible interval
80% credible interval
95% credible interval
Max ECA: 0.14
Median change: −0.5%
ECA

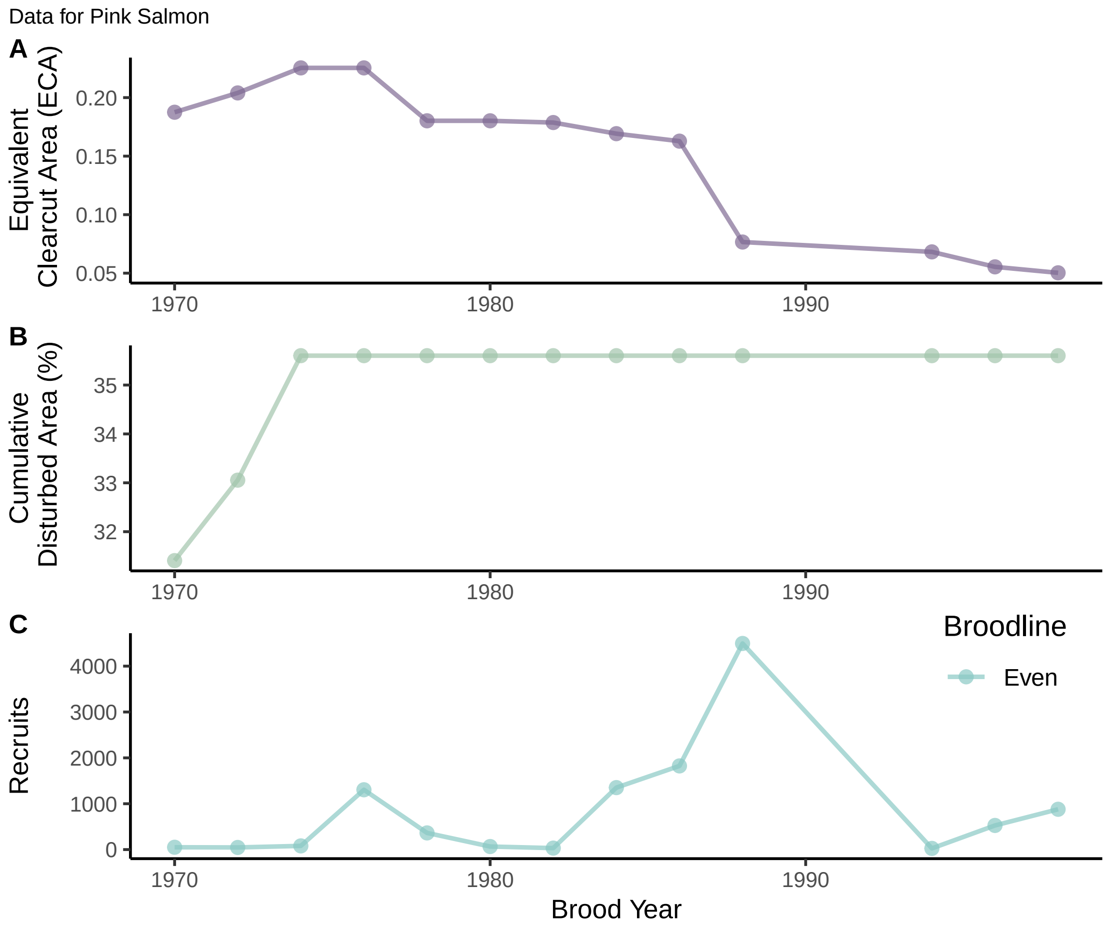
GILFORD CREEK



Model Results for Pink Salmon



MCALISTER CREEK



Model Results for Pink Salmon